

islr-ch7

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Chapter VII - Moving Beyond Linearity

All the questions are as per the [ISL seventh printing of the First edition](#)^{^fn:1}.

Common

```
libsUsed<-c("dplyr","ggplot2","tidyverse",
            "ISLR","caret","MASS", "gridExtra",
            "pls","latex2exp","data.table")
invisible(lapply(libsUsed, library, character.only = TRUE))
```

Question 7.6 - Page 299

In this exercise, you will further analyze the `Wage` data set considered throughout this chapter.

- (a)** Perform polynomial regression to predict `wage` using `age`. Use cross-validation to select the optimal degree d for the polynomial. What degree was chosen, and how does this compare to the results of hypothesis testing using ANOVA? Make a plot of the resulting polynomial fit to the data.
- (b)** Fit a step function to predict `wage` using `age`, and perform cross-validation to choose the optimal number of cuts. Make a plot of the fit obtained. In this exercise, we will generate simulated data, and will then use this data to perform best subset selection.

Answer

Lets get the data.

```
set.seed(1984)
wageDat<-ISLR::Wage
wageDat %>% str %>% print
```

```
## 'data.frame':   3000 obs. of  11 variables:
## $ year       : int   2006 2004 2003 2003 2005 2008 2009 2008 2006 2004 ...
## $ age        : int   18 24 45 43 50 54 44 30 41 52 ...
## $ maritl     : Factor w/ 5 levels "1. Never Married",...: 1 1 2 2 4 2 2 1 1 2 ...
## $ race       : Factor w/ 4 levels "1. White","2. Black",...: 1 1 1 3 1 1 4 3 2 1 ...
## $ education  : Factor w/ 5 levels "1. < HS Grad",...: 1 4 3 4 2 4 3 3 3 2 ...
## $ region     : Factor w/ 9 levels "1. New England",...: 2 2 2 2 2 2 2 2 2 ...
## $ jobclass   : Factor w/ 2 levels "1. Industrial",...: 1 2 1 2 2 2 1 2 2 2 ...
## $ health     : Factor w/ 2 levels "1. <=Good","2. >=Very Good": 1 2 1 2 1 2 2 1 2 2 ...
## $ health_ins : Factor w/ 2 levels "1. Yes","2. No": 2 2 1 1 1 1 1 1 1 1 ...
## $ logwage    : num  4.32 4.26 4.88 5.04 4.32 ...
## $ wage       : num  75 70.5 131 154.7 75 ...
## NULL
```

```
wageDat %>% summary %>% print
```

```
##      year          age          maritl          race
## Min.   :2003   Min.   :18.00   1. Never Married: 648   1. White:2480
## 1st Qu.:2004   1st Qu.:33.75   2. Married   :2074   2. Black: 293
## Median :2006   Median :42.00   3. Widowed   : 19   3. Asian: 190
## Mean   :2006   Mean   :42.41   4. Divorced  : 204   4. Other:  37
## 3rd Qu.:2008   3rd Qu.:51.00   5. Separated : 55
## Max.   :2009   Max.   :80.00
##
##      education          region          jobclass
## 1. < HS Grad      :268   2. Middle Atlantic :3000   1. Industrial :1544
## 2. HS Grad        :971   1. New England  :  0   2. Information:1456
## 3. Some College   :650   3. East North Central:  0
## 4. College Grad   :685   4. West North Central:  0
## 5. Advanced Degree:426   5. South Atlantic   :  0
##                      6. East South Central:  0
##                      (Other)           :  0
##
##      health          health_ins          logwage          wage
## 1. <=Good      : 858   1. Yes:2083   Min.   :3.000   Min.   : 20.09
## 2. >=Very Good:2142   2. No : 917   1st Qu.:4.447   1st Qu.: 85.38
##                      Median :4.653   Median :104.92
##                      Mean   :4.654   Mean   :111.70
##                      3rd Qu.:4.857   3rd Qu.:128.68
##                      Max.   :5.763   Max.   :318.34
##
```

```
wageDat %>% sapply(unique) %>% sapply(length) %>% print
```

```
##      year      age      maritl      race education      region      jobclass
##      7         61         5         4         5         1         2
##      health health_ins      logwage      wage
##      2         2         508         508
```

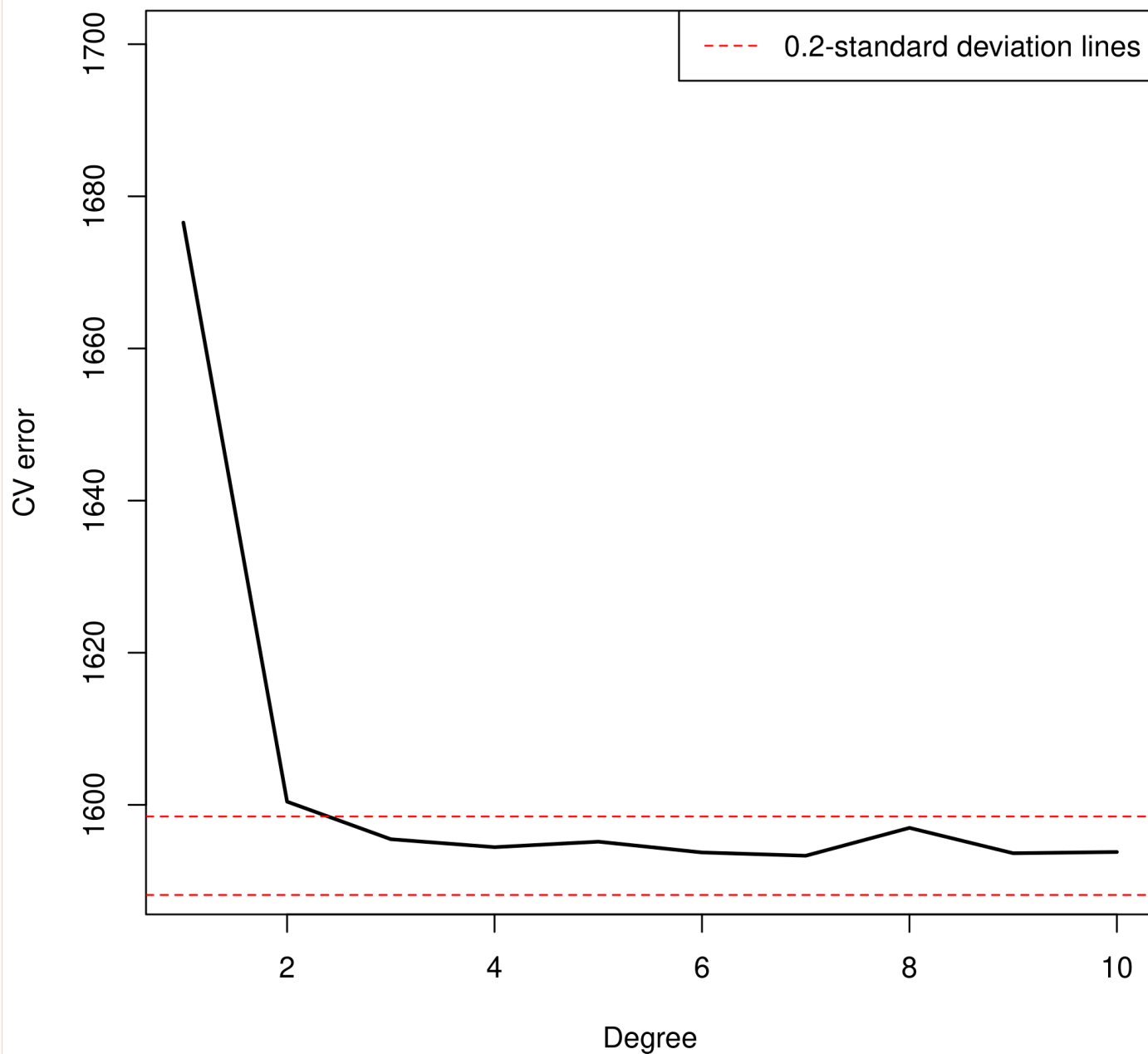
```
library(boot)
```

```
##
## Attaching package: 'boot'
```

```
## The following object is masked from 'package:lattice':
##
##      melanoma
```

a) Polynomial regression

```
all.deltas = rep(NA, 10)
for (i in 1:10) {
  glm.fit = glm(wage~poly(age, i), data=Wage)
  all.deltas[i] = cv.glm(Wage, glm.fit, K=10)$delta[2]
}
plot(1:10, all.deltas, xlab="Degree", ylab="CV error", type="l", pch=20, lwd=2, ylim=c(1590, 1700))
min.point = min(all.deltas)
sd.points = sd(all.deltas)
abline(h=min.point + 0.2 * sd.points, col="red", lty="dashed")
abline(h=min.point - 0.2 * sd.points, col="red", lty="dashed")
legend("topright", "0.2-standard deviation lines", lty="dashed", col="red")
```

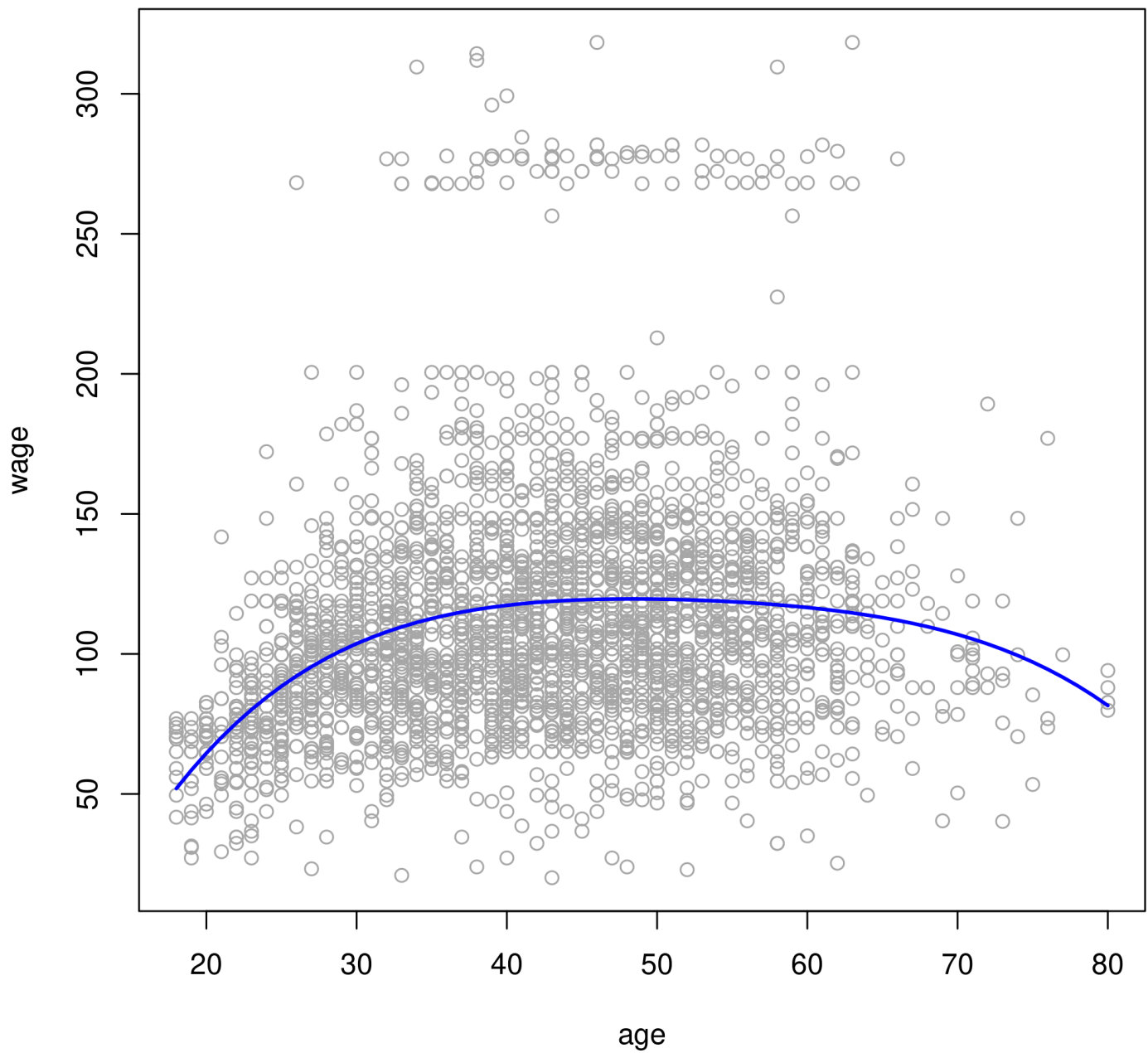


```
# ANOVA
fits=list()
for (i in 1:10) {
  fits[[i]]=glm(wage~poly(age,i),data=wageDat)
}
anova(fits[[1]],fits[[2]],fits[[3]],fits[[4]],fits[[5]],
      fits[[6]],fits[[7]],fits[[8]],fits[[9]],fits[[10]])
```

```
## Analysis of Deviance Table
##
## Model 1: wage ~ poly(age, i)
## Model 2: wage ~ poly(age, i)
## Model 3: wage ~ poly(age, i)
## Model 4: wage ~ poly(age, i)
## Model 5: wage ~ poly(age, i)
## Model 6: wage ~ poly(age, i)
## Model 7: wage ~ poly(age, i)
## Model 8: wage ~ poly(age, i)
## Model 9: wage ~ poly(age, i)
## Model 10: wage ~ poly(age, i)
##      Resid. Df Resid. Dev Df Deviance
## 1          2998      5022216
## 2          2997      4793430  1    228786
## 3          2996      4777674  1    15756
## 4          2995      4771604  1     6070
## 5          2994      4770322  1     1283
## 6          2993      4766389  1     3932
## 7          2992      4763834  1     2555
## 8          2991      4763707  1       127
## 9          2990      4756703  1     7004
## 10         2989      4756701  1         3
```

- The 4th degree looks the best at the moment

```
# 3rd or 4th degrees look best based on ANOVA test
# let's go with 4th degree fit
plot(wage~age, data=wageDat, col="darkgrey")
age.lims = range(wageDat$age)
age.grid = seq(from=age.lims[1], to=age.lims[2])
lm.fit = lm(wage~poly(age, 4), data=wageDat)
lm.pred = predict(lm.fit, data.frame(age=age.grid))
lines(age.grid, lm.pred, col="blue", lwd=2)
```



b) Step function and cross-validation

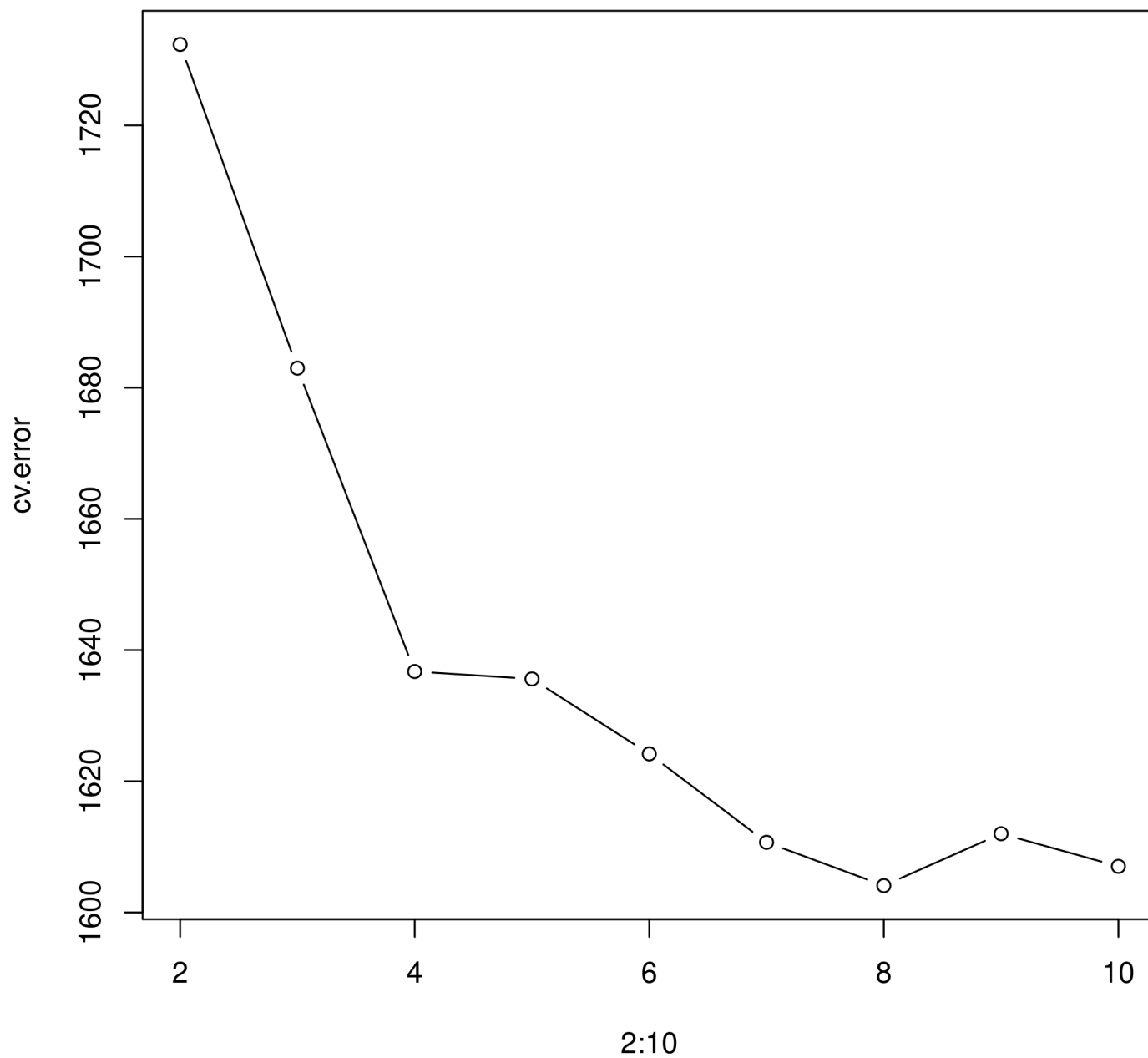
```
# cross-validation
cv.error <- rep(0,9)
for (i in 2:10) {
  wageDat$age.cut <- cut(wageDat$age,i)
  glm.fit <- glm(wage~age.cut, data=wageDat)
  cv.error[i-1] <- cv.glm(wageDat, glm.fit, K=10)$delta[1] # [1]:std, [2]:bias-corrected
}
cv.error
```

```
## [1] 1732.337 1682.978 1636.736 1635.600 1624.174 1610.688 1604.081 1612.005
## [9] 1607.022
```

```
cv.error
```

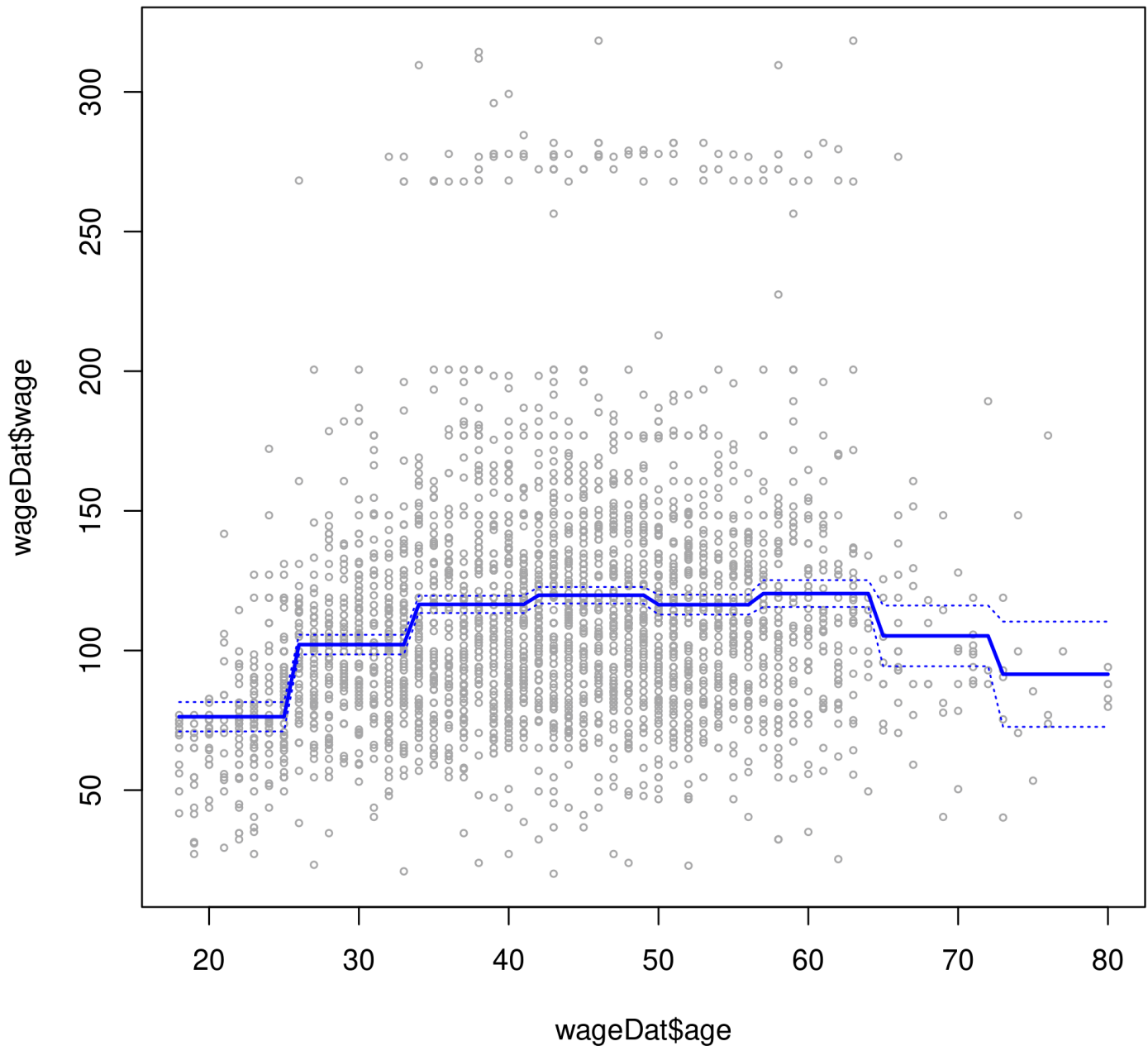
```
## [1] 1732.337 1682.978 1636.736 1635.600 1624.174 1610.688 1604.081 1612.005
## [9] 1607.022
```

```
plot(2:10, cv.error, type="b")
```



```
cut.fit <- glm(wage~cut(age,8), data=wageDat)
preds <- predict(cut.fit, newdata=list(age=age.grid), se=TRUE)
se.bands <- preds$fit + cbind(2*preds$se.fit, -2*preds$se.fit)
plot(wageDat$age, wageDat$wage, xlim=age.lims, cex=0.5, col="darkgrey")
title("Fit with 8 Age Bands")
lines(age.grid, preds$fit, lwd=2, col="blue")
matlines(age.grid, se.bands, lwd=1, col="blue", lty=3)
```

Fit with 8 Age Bands



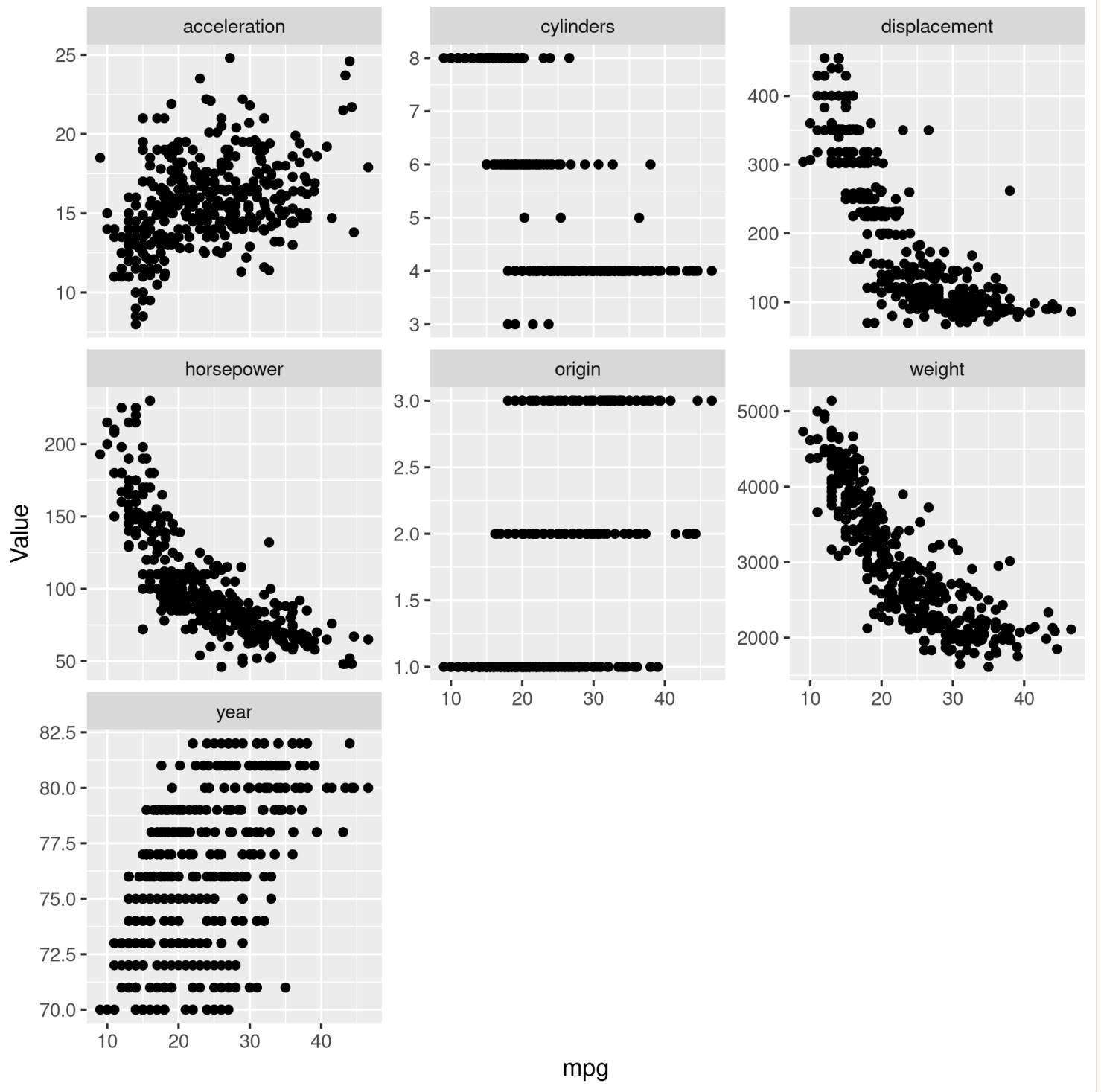
Question 7.8 - Page 299

Fit some of the non-linear models investigated in this chapter to the `Auto` data set. Is there evidence for non-linear relationships in this data set? Create some informative plots to justify your answer.

Answer

```
autoDat<-ISLR::Auto
```

```
autoDat %>% pivot_longer(-c(mpg,name),names_to="Params",values_to="Value") %>% ggplot(aes(x=mpg,y=Value)) +  
  geom_point() +  
  facet_wrap(~ Params, scales = "free_y")
```



Very clearly there is a lot of non-linearity in the `mpg` data, especially for `acceleration`, `weight`, `displacement`, `horsepower`.

```
rss = rep(NA, 10)
fits = list()
for (d in 1:10) {
  fits[[d]] = lm(mpg ~ poly(displacement, d), data = autoDat)
  rss[d] = deviance(fits[[d]])
}
rss %>% print
```

```
## [1] 8378.822 7412.263 7392.322 7391.722 7380.838 7270.746 7089.716 6917.401
## [9] 6737.801 6610.190
```

```
anova(fits[[1]],fits[[2]],fits[[3]],fits[[4]],fits[[5]],
      fits[[6]],fits[[7]],fits[[8]],fits[[9]],fits[[10]])
```



```
## Analysis of Variance Table
##
## Model 1: mpg ~ poly(displacement, d)
## Model 2: mpg ~ poly(displacement, d)
## Model 3: mpg ~ poly(displacement, d)
## Model 4: mpg ~ poly(displacement, d)
## Model 5: mpg ~ poly(displacement, d)
## Model 6: mpg ~ poly(displacement, d)
## Model 7: mpg ~ poly(displacement, d)
## Model 8: mpg ~ poly(displacement, d)
## Model 9: mpg ~ poly(displacement, d)
## Model 10: mpg ~ poly(displacement, d)
##      Res.Df    RSS Df Sum of Sq      F      Pr(>F)
## 1         390 8378.8
## 2         389 7412.3   1    966.56 55.7108 5.756e-13 ***
## 3         388 7392.3   1     19.94  1.1494  0.284364
## 4         387 7391.7   1      0.60  0.0346  0.852549
## 5         386 7380.8   1     10.88  0.6273  0.428823
## 6         385 7270.7   1    110.09  6.3455  0.012177 *
## 7         384 7089.7   1    181.03 10.4343  0.001344 **
## 8         383 6917.4   1    172.31  9.9319  0.001753 **
## 9         382 6737.8   1    179.60 10.3518  0.001404 **
## 10        381 6610.2   1    127.61  7.3553  0.006990 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Confirming our visual indications, we see that the second degree models work well.

```
library(glmnet)
```

```
## Loading required package: Matrix
```

```
##
## Attaching package: 'Matrix'
```

```
## The following objects are masked from 'package:tidyr':
##
##      expand, pack, unpack
```

```
## Loaded glmnet 3.0-2
```

```
library(boot)
```

```
cv.errs = rep(NA, 15)
for (d in 1:15) {
  fit = glm(mpg ~ poly(displacement, d), data = Auto)
  cv.errs[d] = cv.glm(Auto, fit, K = 15)$delta[2]
}
which.min(cv.errs)
```

```
## [1] 10
```

Strangely, we seem to have ended up with a ten variable model here.

```
# Step functions
cv.errs = rep(NA, 10)
for (c in 2:10) {
  Auto$dis.cut = cut(Auto$displacement, c)
  fit = glm(mpg ~ dis.cut, data = Auto)
  cv.errs[c] = cv.glm(Auto, fit, K = 10)$delta[2]
}
which.min(cv.errs) %>% print
```

```
## [1] 9
```

```
library(splines)
cv.errs = rep(NA, 10)
for (df in 3:10) {
  fit = glm(mpg ~ ns(displacement, df = df), data = Auto)
  cv.errs[df] = cv.glm(Auto, fit, K = 10)$delta[2]
}
which.min(cv.errs) %>% print
```

```
## [1] 10
```

```
library(gam)
```

```
## Loading required package: foreach
```

```
##
## Attaching package: 'foreach'
```

```
## The following objects are masked from 'package:purrr':
##
##   accumulate, when
```

```
## Loaded gam 1.16.1
```

```
# GAMs
fit = gam(mpg ~ s(displacement, 4) + s(horsepower, 4), data = Auto)
```

```
## Warning in model.matrix.default(mt, mf, contrasts): non-list contrasts argument
## ignored
```

```
summary(fit)
```

```
##
## Call: gam(formula = mpg ~ s(displacement, 4) + s(horsepower, 4), data = Auto)
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -11.2982  -2.1592  -0.4394   2.1247  17.0946
##
## (Dispersion Parameter for gaussian family taken to be 15.3543)
##
##      Null Deviance: 23818.99 on 391 degrees of freedom
## Residual Deviance: 5880.697 on 382.9999 degrees of freedom
## AIC: 2194.05
##
## Number of Local Scoring Iterations: 2
##
## Anova for Parametric Effects
##              Df Sum Sq Mean Sq F value    Pr(>F)
## s(displacement, 4)    1 15254.9 15254.9 993.524 < 2e-16 ***
## s(horsepower, 4)      1  1038.4   1038.4  67.632 3.1e-15 ***
## Residuals           383   5880.7     15.4
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Anova for Nonparametric Effects
##              Npar Df Npar F      Pr(F)
## (Intercept)
## s(displacement, 4)      3 13.613 1.863e-08 ***
## s(horsepower, 4)       3 15.606 1.349e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Question 7.9 - Pages 299-300

This question uses the variables `dis` (the weighted mean of distances to five `Boston` employment centers) and `nox` (nitrogen oxides concentration in parts per 10 million) from the `Boston` data. We will treat `dis` as the predictor and `nox` as the response.

- (a)** Use the `poly()` function to fit a cubic polynomial regression to predict `nox` using `dis`. Report the regression output, and plot the resulting data and polynomial fits.
- (b)** Plot the polynomial fits for a range of different polynomial degrees (say, from 1 to 10), and report the associated residual sum of squares.
- (c)** Perform cross-validation or another approach to select the optimal degree for the polynomial, and explain your results.
- (d)** Use the `bs()` function to fit a regression spline to predict `nox` using `dis`. Report the output for the fit using four degrees of freedom. How did you choose the knots? Plot the resulting fit.
- (e)** Now fit a regression spline for a range of degrees of freedom, and plot the resulting fits and report the resulting RSS. Describe the results obtained.
- (f)** Perform cross-validation or another approach in order to select the best degrees of freedom for a regression spline on this data. Describe your results.

Answer

```
boston<-MASS::Boston
boston %>% str %>% print
```

```
## 'data.frame':   506 obs. of  14 variables:
##  $ crim   : num  0.00632 0.02731 0.02729 0.03237 0.06905 ...
##  $ zn     : num  18 0 0 0 0 12.5 12.5 12.5 12.5 ...
##  $ indus  : num  2.31 7.07 7.07 2.18 2.18 2.18 7.87 7.87 7.87 ...
##  $ chas   : int   0 0 0 0 0 0 0 0 0 ...
##  $ nox    : num  0.538 0.469 0.469 0.458 0.458 0.458 0.524 0.524 0.524 ...
##  $ rm     : num  6.58 6.42 7.18 7 7.15 ...
##  $ age    : num  65.2 78.9 61.1 45.8 54.2 58.7 66.6 96.1 100 85.9 ...
##  $ dis    : num  4.09 4.97 4.97 6.06 6.06 ...
##  $ rad    : int   1 2 2 3 3 3 5 5 5 ...
##  $ tax    : num  296 242 242 222 222 222 311 311 311 ...
##  $ ptratio: num  15.3 17.8 17.8 18.7 18.7 18.7 15.2 15.2 15.2 ...
##  $ black  : num  397 397 393 395 397 ...
##  $ lstat  : num  4.98 9.14 4.03 2.94 5.33 ...
##  $ medv   : num  24 21.6 34.7 33.4 36.2 28.7 22.9 27.1 16.5 18.9 ...
## NULL
```

```
boston %>% summary %>% print
```

```
##      crim      zn      indus      chas
## Min.   : 0.00632   Min.   : 0.00   Min.   : 0.46   Min.   :0.00000
## 1st Qu.: 0.08204   1st Qu.: 0.00   1st Qu.: 5.19   1st Qu.:0.00000
## Median : 0.25651   Median : 0.00   Median : 9.69   Median :0.00000
## Mean   : 3.61352   Mean   : 11.36   Mean   :11.14   Mean   :0.06917
## 3rd Qu.: 3.67708   3rd Qu.: 12.50   3rd Qu.:18.10   3rd Qu.:0.00000
## Max.   :88.97620   Max.   :100.00   Max.   :27.74   Max.   :1.00000
##      nox      rm      age      dis
## Min.   :0.3850   Min.   :3.561   Min.   : 2.90   Min.   : 1.130
## 1st Qu.:0.4490   1st Qu.:5.886   1st Qu.: 45.02   1st Qu.: 2.100
## Median :0.5380   Median :6.208   Median : 77.50   Median : 3.207
## Mean   :0.5547   Mean   :6.285   Mean   : 68.57   Mean   : 3.795
## 3rd Qu.:0.6240   3rd Qu.:6.623   3rd Qu.: 94.08   3rd Qu.: 5.188
## Max.   :0.8710   Max.   :8.780   Max.   :100.00   Max.   :12.127
##      rad      tax      ptratio      black
## Min.   : 1.000   Min.   :187.0   Min.   :12.60   Min.   : 0.32
## 1st Qu.: 4.000   1st Qu.:279.0   1st Qu.:17.40   1st Qu.:375.38
## Median : 5.000   Median :330.0   Median :19.05   Median :391.44
## Mean   : 9.549   Mean   :408.2   Mean   :18.46   Mean   :356.67
## 3rd Qu.:24.000   3rd Qu.:666.0   3rd Qu.:20.20   3rd Qu.:396.23
## Max.   :24.000   Max.   :711.0   Max.   :22.00   Max.   :396.90
##      lstat      medv
## Min.   : 1.73   Min.   : 5.00
## 1st Qu.: 6.95   1st Qu.:17.02
## Median :11.36   Median :21.20
## Mean   :12.65   Mean   :22.53
## 3rd Qu.:16.95   3rd Qu.:25.00
## Max.   :37.97   Max.   :50.00
```

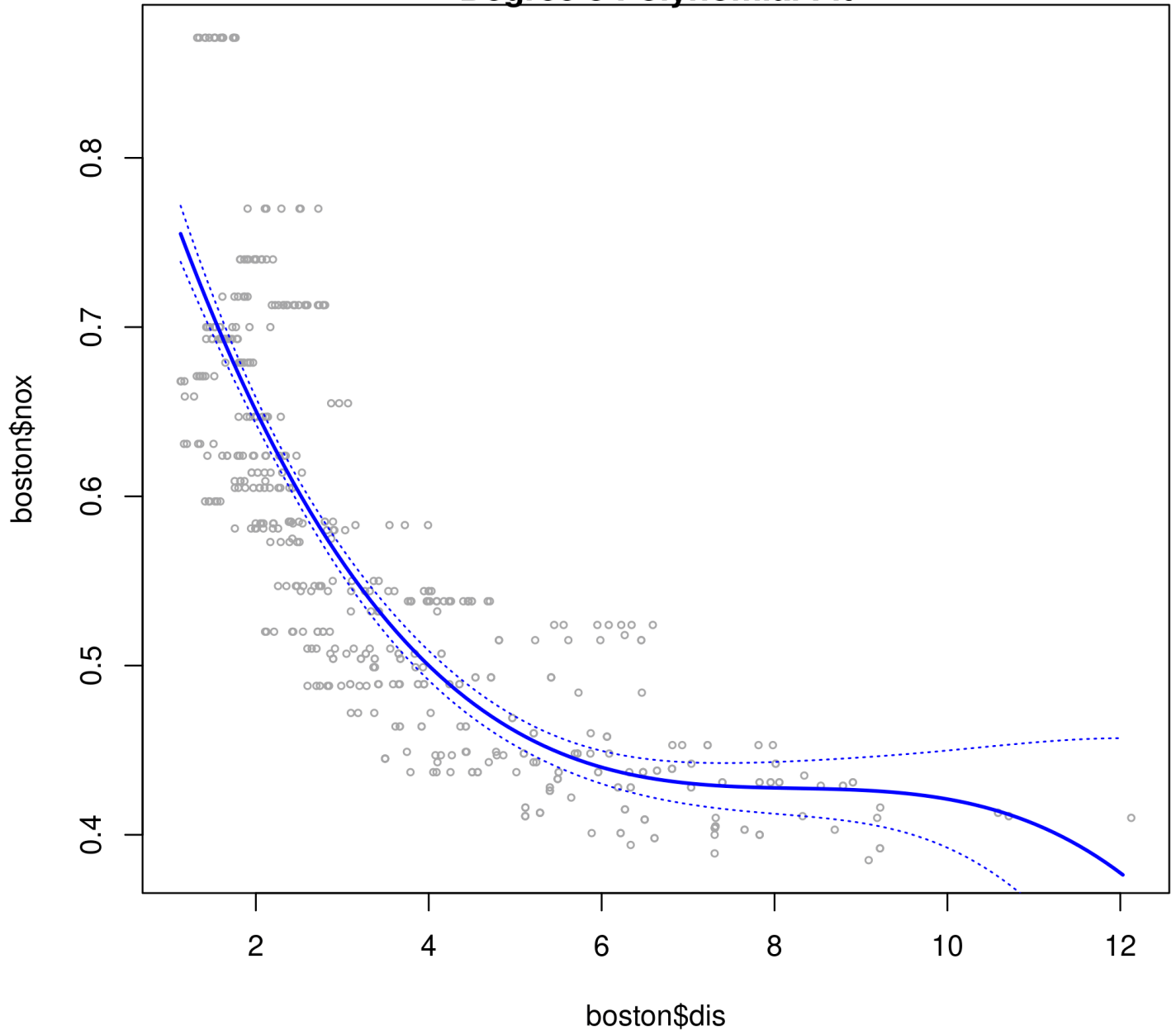
```
boston %>% sapply(unique) %>% sapply(length) %>% print
```

```
##      crim      zn      indus      chas      nox      rm      age      dis      rad      tax
##      504      26      76      2      81      446      356      412      9      66
## ptratio      black      lstat      medv
##      46      357      455      229
```

a) Polynomial

```
fit.03 <- lm(nox~poly(dis,3), data=boston)
dislims <- range(boston$dis)
dis.grid <- seq(dislims[1], dislims[2], 0.1)
preds <- predict(fit.03, newdata=list(dis=dis.grid), se=TRUE)
se.bands <- preds$fit + cbind(2*preds$se.fit, -2*preds$se.fit)
par(mfrow=c(1,1), mar=c(4.5,4.5,1,1), oma=c(0,0,4,0))
plot(boston$dis, boston$nox, xlim=dislims, cex=0.5, col="darkgrey")
title("Degree 3 Polynomial Fit")
lines(dis.grid, preds$fit, lwd=2, col="blue")
matlines(dis.grid, se.bands, lwd=1, col="blue", lty=3)
```

Degree 3 Polynomial Fit



```
summary(fit.03)
```

```
##
## Call:
## lm(formula = nox ~ poly(dis, 3), data = boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.121130 -0.040619 -0.009738  0.023385  0.194904
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.554695   0.002759  201.021 < 2e-16 ***
## poly(dis, 3)1 -2.003096   0.062071 -32.271 < 2e-16 ***
## poly(dis, 3)2  0.856330   0.062071  13.796 < 2e-16 ***
## poly(dis, 3)3 -0.318049   0.062071  -5.124 4.27e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.06207 on 502 degrees of freedom
## Multiple R-squared:  0.7148, Adjusted R-squared:  0.7131
## F-statistic: 419.3 on 3 and 502 DF,  p-value: < 2.2e-16
```

b) Multiple Polynomials

```

rss.error <- rep(0,10)
for (i in 1:10) {
  lm.fit <- lm(nox~poly(dis,i), data=boston)
  rss.error[i] <- sum(lm.fit$residuals^2)
}
rss.error

```

```

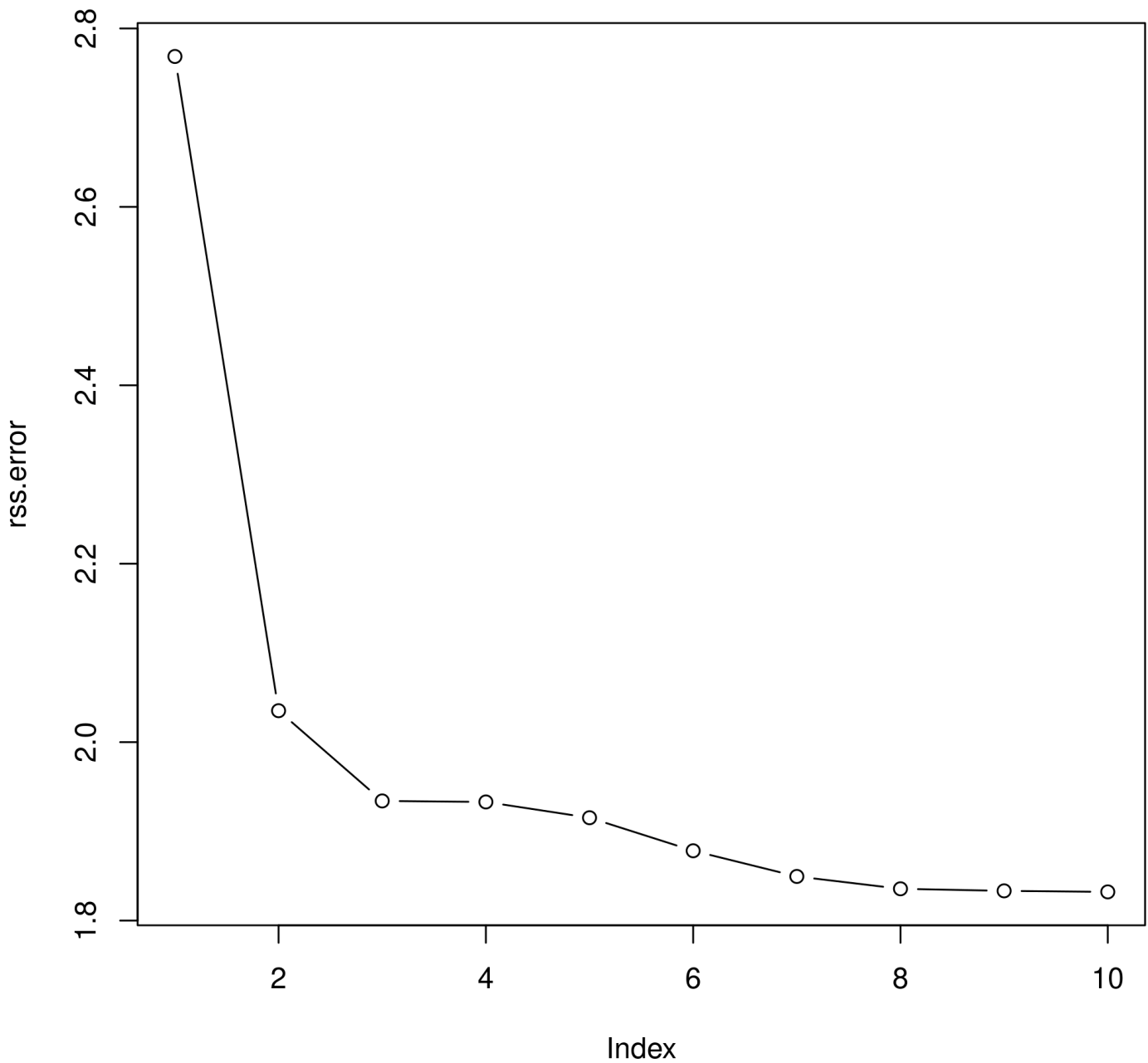
## [1] 2.768563 2.035262 1.934107 1.932981 1.915290 1.878257 1.849484 1.835630
## [9] 1.833331 1.832171

```

```

plot(rss.error, type="b")

```



c) Cross validation and polynomial selection

```

require(boot)
set.seed(1)
cv.error <- rep(0,10)
for (i in 1:10) {
  glm.fit <- glm(nox~poly(dis,i), data=boston)
  cv.error[i] <- cv.glm(boston, glm.fit, K=10)$delta[1] # [1]:std, [2]:bias-corrected
}
cv.error

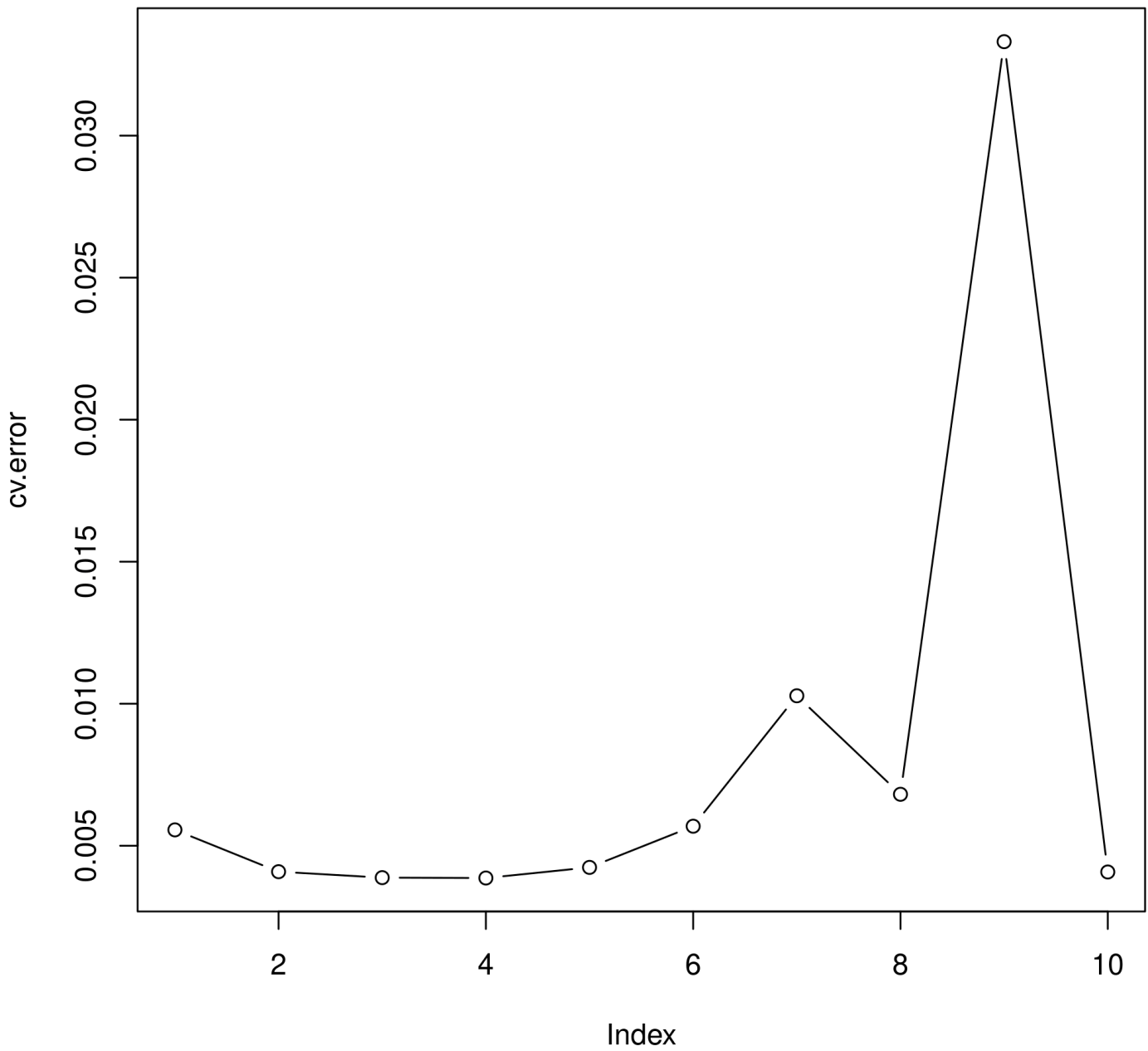
```

```

## [1] 0.005558263 0.004085706 0.003876521 0.003863342 0.004237452 0.005686862
## [7] 0.010278897 0.006810868 0.033308607 0.004075599

```

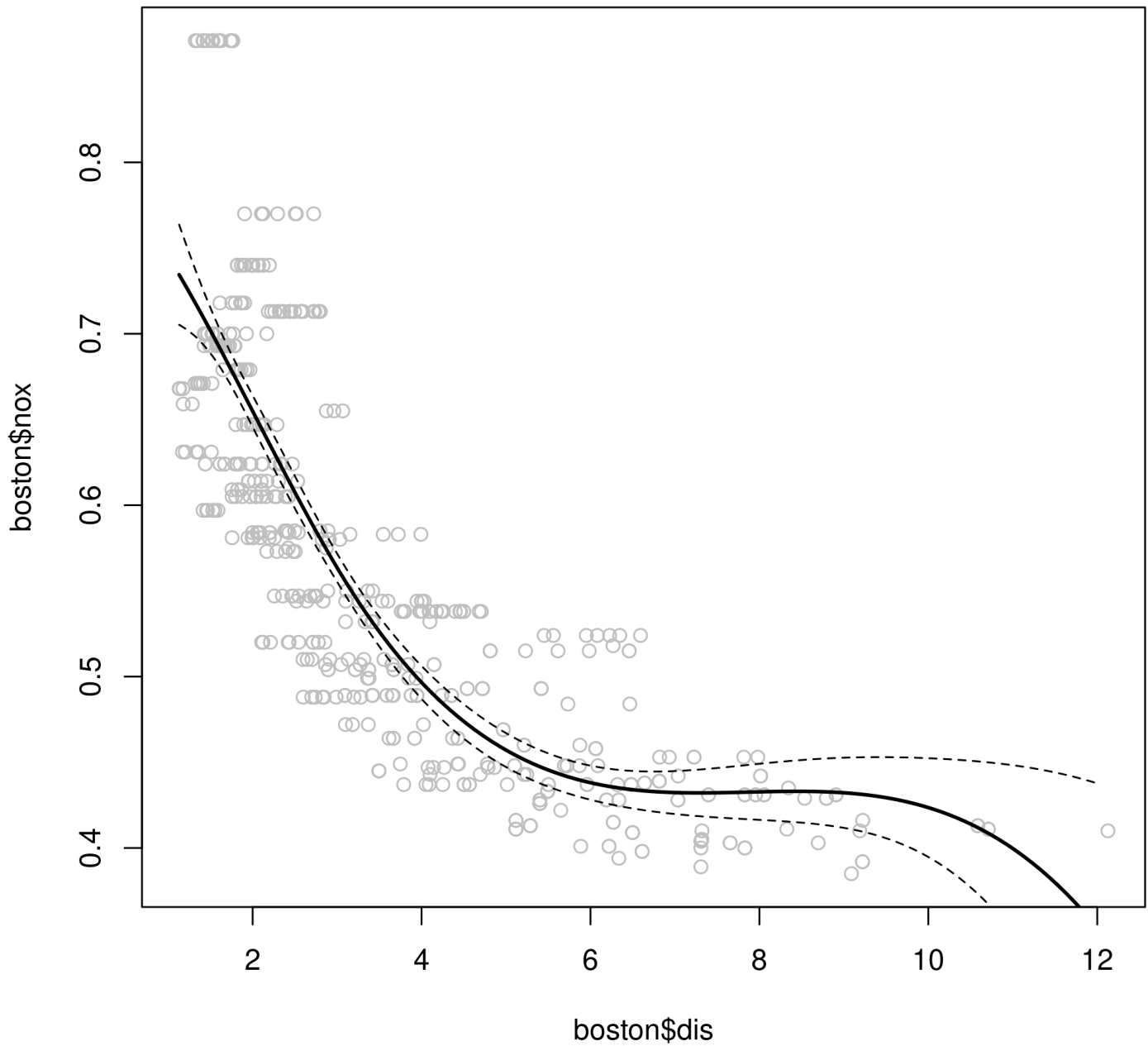
```
plot(cv.error, type="b")
```



- I feel like the second degree fit would be the most reasonable, though the fourth degree seems to be doing well.

d) Regression spline

```
fit.sp <- lm(nox~bs(dis, df=4), data=boston)
pred <- predict(fit.sp, newdata=list(dis=dis.grid), se=T)
plot(boston$dis, boston$nox, col="gray")
lines(dis.grid, pred$fit, lwd=2)
lines(dis.grid, pred$fit+2*pred$se, lty="dashed")
lines(dis.grid, pred$fit-2*pred$se, lty="dashed")
```



```
# set df to select knots at uniform quantiles of `dis`
attr(bs(boston$dis,df=4),"knots") # only 1 knot at 50th percentile
```

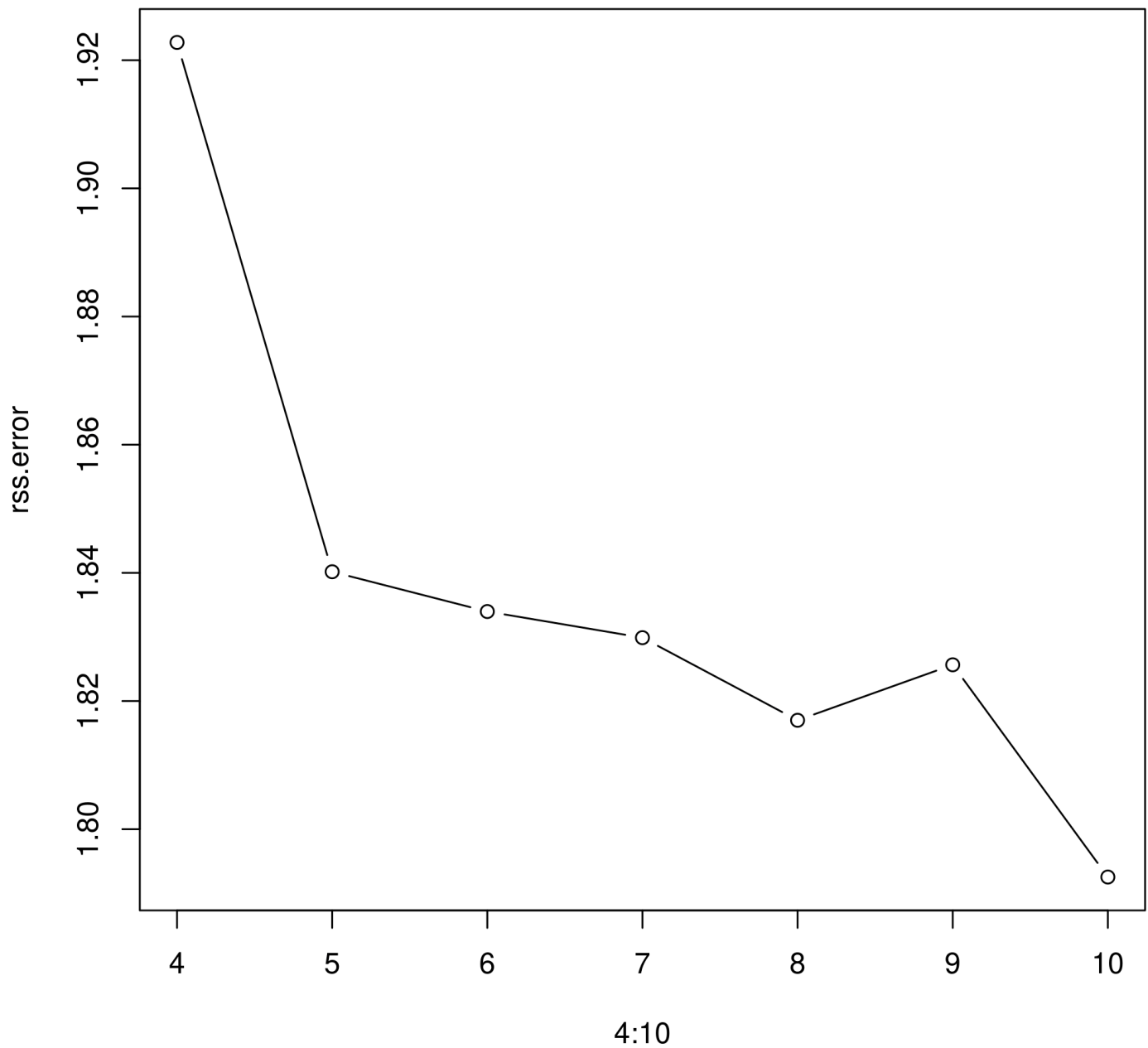
```
##      50%
## 3.20745
```

e) Range of regression splines

```
rss.error <- rep(0,7)
for (i in 4:10) {
  fit.sp <- lm(nox~bs(dis, df=i), data=boston)
  rss.error[i-3] <- sum(fit.sp$residuals^2)
}
rss.error
```

```
## [1] 1.922775 1.840173 1.833966 1.829884 1.816995 1.825653 1.792535
```

```
plot(4:10, rss.error, type="b")
```



- As the model gains more degrees of freedom, it tends to over fit to the training data better

f) Cross validation for best spline

```
cv.error <- rep(0,7)
for (i in 4:10) {
  glm.fit <- glm(nox~bs(dis, df=i), data=boston)
  cv.error[i-3] <- cv.glm(boston, glm.fit, K=10)$delta[1]
}
```

```
## Warning in bs(dis, degree = 3L, knots = c(`50%` = 3.1523), Boundary.knots =
## c(1.1296, : some 'x' values beyond boundary knots may cause ill-conditioned
## bases
```

```
## Warning in bs(dis, degree = 3L, knots = c(`50%` = 3.1523), Boundary.knots =
## c(1.1296, : some 'x' values beyond boundary knots may cause ill-conditioned
## bases
```

```
## Warning in bs(dis, degree = 3L, knots = c(`50%` = 3.2157), Boundary.knots =
## c(1.137, : some 'x' values beyond boundary knots may cause ill-conditioned bases
```

```
## Warning in bs(dis, degree = 3L, knots = c(`50%` = 3.2157), Boundary.knots =
## c(1.137, : some 'x' values beyond boundary knots may cause ill-conditioned bases
```



```
## Warning in bs(dis, degree = 3L, knots = c(`33.33333%` = 2.35953333333333, : some
## 'x' values beyond boundary knots may cause ill-conditioned bases

## Warning in bs(dis, degree = 3L, knots = c(`33.33333%` = 2.35953333333333, : some
## 'x' values beyond boundary knots may cause ill-conditioned bases
```

```
## Warning in bs(dis, degree = 3L, knots = c(`33.33333%` = 2.38403333333333, : some
## 'x' values beyond boundary knots may cause ill-conditioned bases

## Warning in bs(dis, degree = 3L, knots = c(`33.33333%` = 2.38403333333333, : some
## 'x' values beyond boundary knots may cause ill-conditioned bases
```

```
## Warning in bs(dis, degree = 3L, knots = c(`25%` = 2.07945, `50%` = 3.1323, :
## some 'x' values beyond boundary knots may cause ill-conditioned bases

## Warning in bs(dis, degree = 3L, knots = c(`25%` = 2.07945, `50%` = 3.1323, :
## some 'x' values beyond boundary knots may cause ill-conditioned bases
```

```
## Warning in bs(dis, degree = 3L, knots = c(`25%` = 2.1103, `50%` = 3.2797, : some
## 'x' values beyond boundary knots may cause ill-conditioned bases

## Warning in bs(dis, degree = 3L, knots = c(`25%` = 2.1103, `50%` = 3.2797, : some
## 'x' values beyond boundary knots may cause ill-conditioned bases
```

```
## Warning in bs(dis, degree = 3L, knots = c(`20%` = 1.9682, `40%` = 2.7147, : some
## 'x' values beyond boundary knots may cause ill-conditioned bases

## Warning in bs(dis, degree = 3L, knots = c(`20%` = 1.9682, `40%` = 2.7147, : some
## 'x' values beyond boundary knots may cause ill-conditioned bases
```

```
## Warning in bs(dis, degree = 3L, knots = c(`20%` = 1.95434, `40%` = 2.59666, :
## some 'x' values beyond boundary knots may cause ill-conditioned bases

## Warning in bs(dis, degree = 3L, knots = c(`20%` = 1.95434, `40%` = 2.59666, :
## some 'x' values beyond boundary knots may cause ill-conditioned bases
```

```
## Warning in bs(dis, degree = 3L, knots = c(`16.66667%` = 1.82203333333333, : some
## 'x' values beyond boundary knots may cause ill-conditioned bases

## Warning in bs(dis, degree = 3L, knots = c(`16.66667%` = 1.82203333333333, : some
## 'x' values beyond boundary knots may cause ill-conditioned bases
```

```
## Warning in bs(dis, degree = 3L, knots = c(`16.66667%` = 1.8226, `33.33333%` =
## 2.3817, : some 'x' values beyond boundary knots may cause ill-conditioned bases

## Warning in bs(dis, degree = 3L, knots = c(`16.66667%` = 1.8226, `33.33333%` =
## 2.3817, : some 'x' values beyond boundary knots may cause ill-conditioned bases
```

```
## Warning in bs(dis, degree = 3L, knots = c(`14.28571%` = 1.7936, `28.57143%`
## = 2.16972857142857, : some 'x' values beyond boundary knots may cause ill-
## conditioned bases

## Warning in bs(dis, degree = 3L, knots = c(`14.28571%` = 1.7936, `28.57143%`
## = 2.16972857142857, : some 'x' values beyond boundary knots may cause ill-
## conditioned bases
```

```
## Warning in bs(dis, degree = 3L, knots = c(`12.5%` = 1.754625, `25%` = 2.10215, :
## some 'x' values beyond boundary knots may cause ill-conditioned bases

## Warning in bs(dis, degree = 3L, knots = c(`12.5%` = 1.754625, `25%` = 2.10215, :
## some 'x' values beyond boundary knots may cause ill-conditioned bases
```

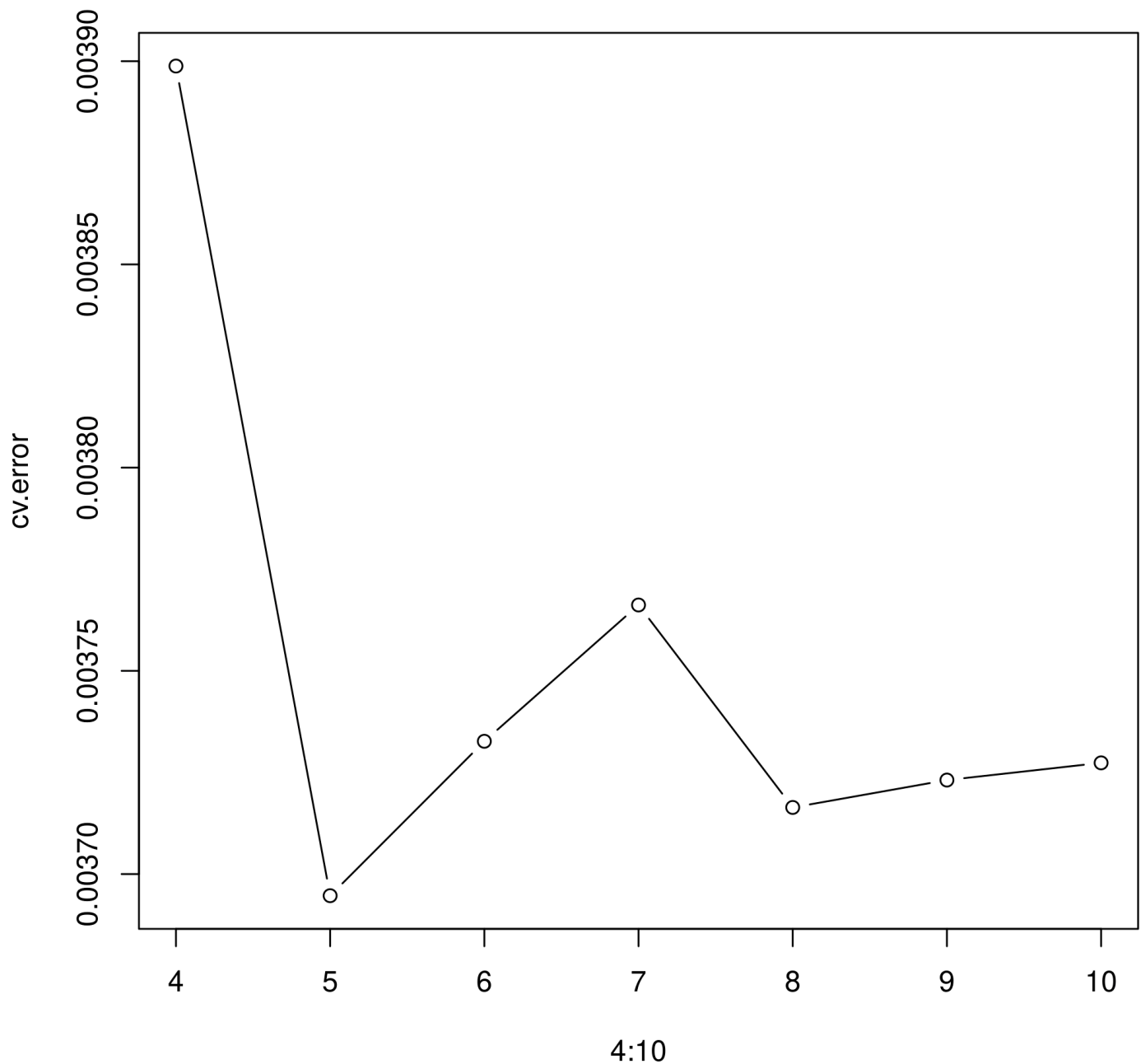
```
## Warning in bs(dis, degree = 3L, knots = c(`12.5%` = 1.751575, `25%` = 2.08755, :
## some 'x' values beyond boundary knots may cause ill-conditioned bases

## Warning in bs(dis, degree = 3L, knots = c(`12.5%` = 1.751575, `25%` = 2.08755, :
## some 'x' values beyond boundary knots may cause ill-conditioned bases
```

```
cv.error
```

```
## [1] 0.003898810 0.003694675 0.003732665 0.003766202 0.003716389 0.003723126
## [7] 0.003727358
```

```
plot(4:10, cv.error, type="b")
```



- A fifth degree polynomial is clearly indicated

Question 10 - Page 300

This question relates to the `College` data set.

- Split the data into a training set and a test set. Using out-of-state tuition as the response and the other variables as the predictors, perform forward stepwise selection on the training set in order to identify a satisfactory model that uses just a subset of the predictors.
- Fit a GAM on the training data, using out-of-state tuition as the response and the features selected in the previous step as the predictors. Plot the results, and explain your findings.
- Evaluate the model obtained on the test set, and explain the results obtained.
- For which variables, if any, is there evidence of a non-linear relationship with the response?

Answer

```
colDat<-ISLR::College
colDat %>% str %>% print
```

```
## 'data.frame':   777 obs. of  18 variables:
## $ Private      : Factor w/ 2 levels "No","Yes": 2 2 2 2 2 2 2 2 2 ...
## $ Apps         : num  1660 2186 1428 417 193 ...
## $ Accept       : num  1232 1924 1097 349 146 ...
## $ Enroll       : num  721 512 336 137 55 158 103 489 227 172 ...
## $ Top10perc    : num   23 16 22 60 16 38 17 37 30 21 ...
## $ Top25perc    : num   52 29 50 89 44 62 45 68 63 44 ...
## $ F.Undergrad  : num  2885 2683 1036 510 249 ...
## $ P.Undergrad  : num   537 1227 99 63 869 ...
## $ Outstate     : num  7440 12280 11250 12960 7560 ...
## $ Room.Board   : num  3300 6450 3750 5450 4120 ...
## $ Books        : num   450 750 400 450 800 500 500 450 300 660 ...
## $ Personal     : num  2200 1500 1165 875 1500 ...
## $ PhD          : num   70 29 53 92 76 67 90 89 79 40 ...
## $ Terminal     : num   78 30 66 97 72 73 93 100 84 41 ...
## $ S.F.Ratio    : num  18.1 12.2 12.9 7.7 11.9 9.4 11.5 13.7 11.3 11.5 ...
## $ perc.alumni  : num   12 16 30 37 2 11 26 37 23 15 ...
## $ Expend       : num  7041 10527 8735 19016 10922 ...
## $ Grad.Rate    : num   60 56 54 59 15 55 63 73 80 52 ...
## NULL
```

```
colDat %>% summary %>% print
```

```
## Private      Apps      Accept      Enroll      Top10perc
## No :212      Min.   : 81      Min.   : 72      Min.   : 35      Min.   : 1.00
## Yes:565      1st Qu.: 776      1st Qu.: 604      1st Qu.: 242      1st Qu.:15.00
##           Median : 1558      Median : 1110      Median : 434      Median :23.00
##           Mean   : 3002      Mean   : 2019      Mean   : 780      Mean   :27.56
##           3rd Qu.: 3624      3rd Qu.: 2424      3rd Qu.: 902      3rd Qu.:35.00
##           Max.   :48094      Max.   :26330      Max.   :6392      Max.   :96.00
## Top25perc    F.Undergrad  P.Undergrad      Outstate
## Min.   : 9.0      Min.   : 139      Min.   : 1.0      Min.   : 2340
## 1st Qu.: 41.0      1st Qu.: 992      1st Qu.: 95.0      1st Qu.: 7320
## Median : 54.0      Median : 1707      Median : 353.0      Median : 9990
## Mean   : 55.8      Mean   : 3700      Mean   : 855.3      Mean   :10441
## 3rd Qu.: 69.0      3rd Qu.: 4005      3rd Qu.: 967.0      3rd Qu.:12925
## Max.   :100.0      Max.   :31643      Max.   :21836.0      Max.   :21700
## Room.Board   Books      Personal      PhD
## Min.   :1780      Min.   : 96.0      Min.   : 250      Min.   : 8.00
## 1st Qu.:3597      1st Qu.: 470.0      1st Qu.: 850      1st Qu.: 62.00
## Median :4200      Median : 500.0      Median :1200      Median : 75.00
## Mean   :4358      Mean   : 549.4      Mean   :1341      Mean   : 72.66
## 3rd Qu.:5050      3rd Qu.: 600.0      3rd Qu.:1700      3rd Qu.: 85.00
## Max.   :8124      Max.   :2340.0      Max.   :6800      Max.   :103.00
## Terminal     S.F.Ratio    perc.alumni      Expend
## Min.   : 24.0      Min.   : 2.50      Min.   : 0.00      Min.   : 3186
## 1st Qu.: 71.0      1st Qu.:11.50      1st Qu.:13.00      1st Qu.: 6751
## Median : 82.0      Median :13.60      Median :21.00      Median : 8377
## Mean   : 79.7      Mean   :14.09      Mean   :22.74      Mean   : 9660
## 3rd Qu.: 92.0      3rd Qu.:16.50      3rd Qu.:31.00      3rd Qu.:10830
## Max.   :100.0      Max.   :39.80      Max.   :64.00      Max.   :56233
## Grad.Rate
## Min.   : 10.00
## 1st Qu.: 53.00
## Median : 65.00
## Mean   : 65.46
## 3rd Qu.: 78.00
## Max.   :118.00
```

```
colDat %>% sapply(unique) %>% sapply(length) %>% print
```

```
## Private      Apps      Accept      Enroll      Top10perc      Top25perc
## 2           711           693           581           82           89
## F.Undergrad P.Undergrad      Outstate      Room.Board      Books      Personal
## 714           566           640           553           122           294
## PhD          Terminal     S.F.Ratio    perc.alumni      Expend      Grad.Rate
## 78           65           173           61           744           81
```

```
plotLEAP=function(leapObj){
  par(mfrow = c(2,2))
  bar2=which.max(leapObj$adjr2)
  bbic=which.min(leapObj$bic)
  bcp=which.min(leapObj$cp)
  plot(leapObj$rss,xlab="Number of variables",ylab="RSS",type="b")
  plot(leapObj$adjr2,xlab="Number of variables",ylab=TeX("Adjusted R^2"),type="b")
  points(bar2, leapObj$adjr2[bar2],col="green",cex=2,pch=20)
  plot(leapObj$bic,xlab="Number of variables",ylab=TeX("BIC"),type="b")
  points(bbic, leapObj$bic[bbic],col="blue",cex=2,pch=20)
  plot(leapObj$cp,xlab="Number of variables",ylab=TeX("C_p"),type="b")
  points(bcp, leapObj$cp[bcp],col="red",cex=2,pch=20)
}
```

a) Train test

```
train_ind = sample(colDat %>% nrow,100)
test_ind = setdiff(seq_len(colDat %>% nrow), train_ind)
```

Best subset selection

```
train_set<-colDat[train_ind,]
test_set<-colDat[-train_ind,]
```

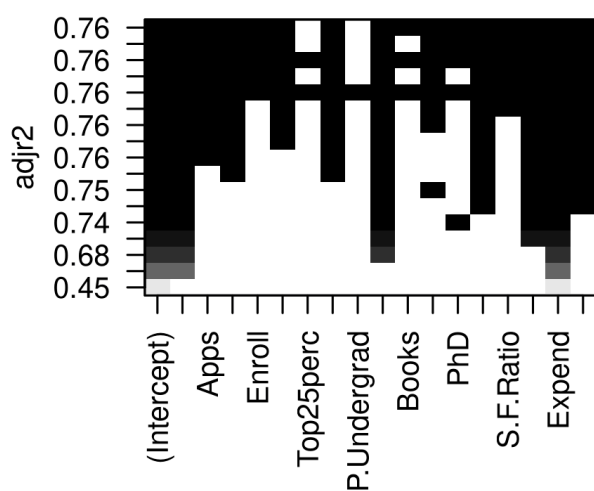
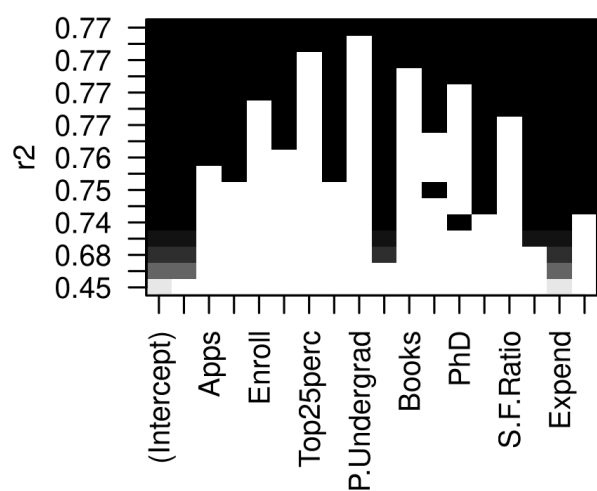
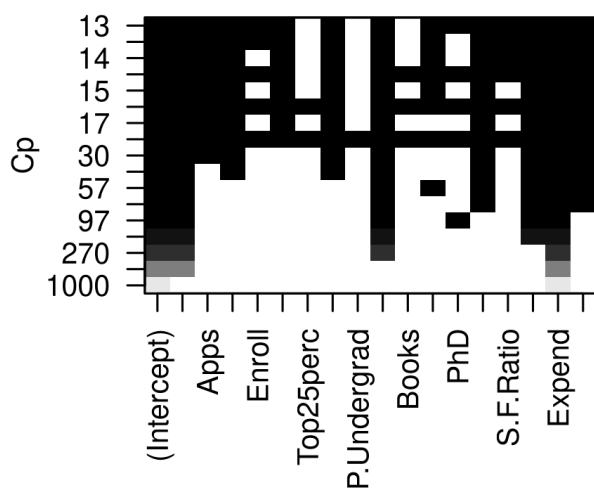
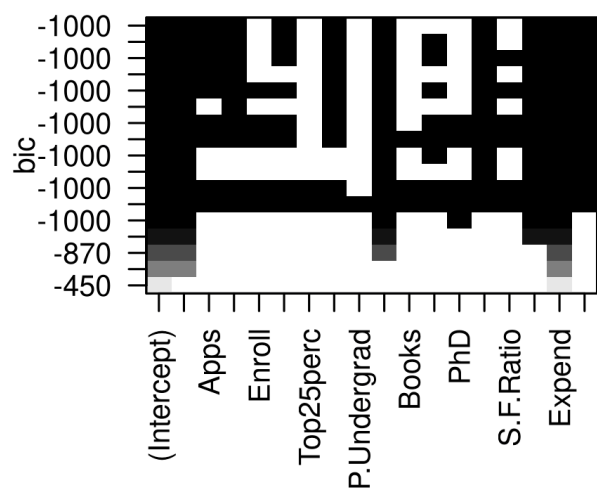
```
library(leaps)
```

```
modelFit<-regsubsets(Outstate~.,data=colDat,nvmax=20)
modelFit %>% summary %>% print
```

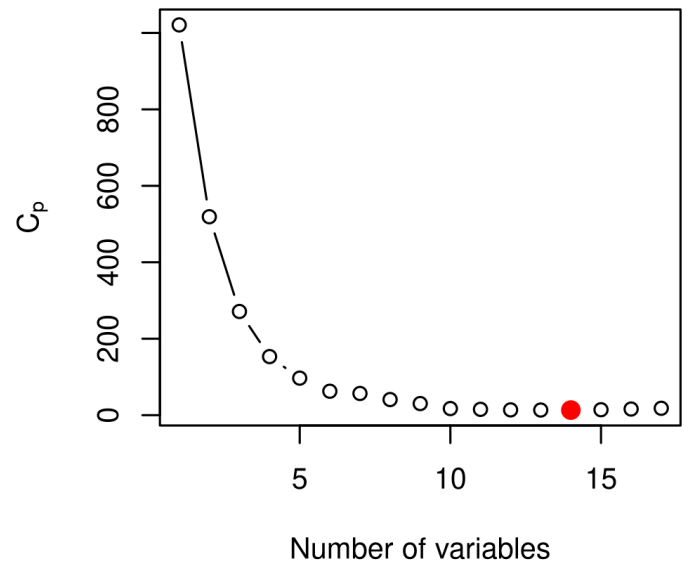
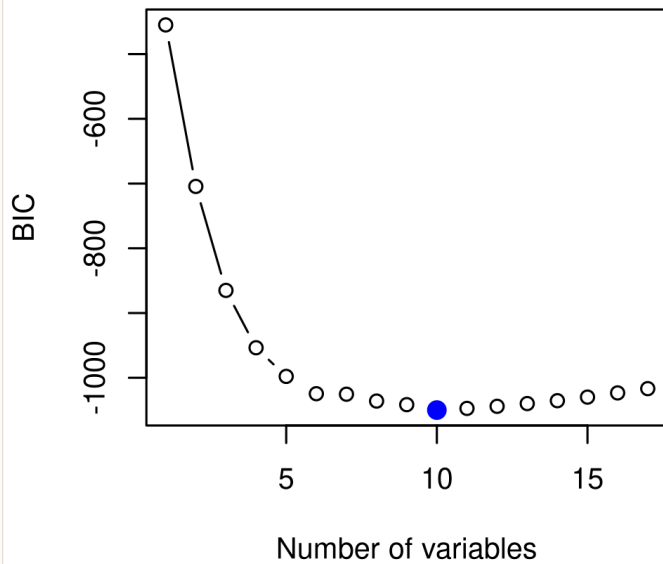
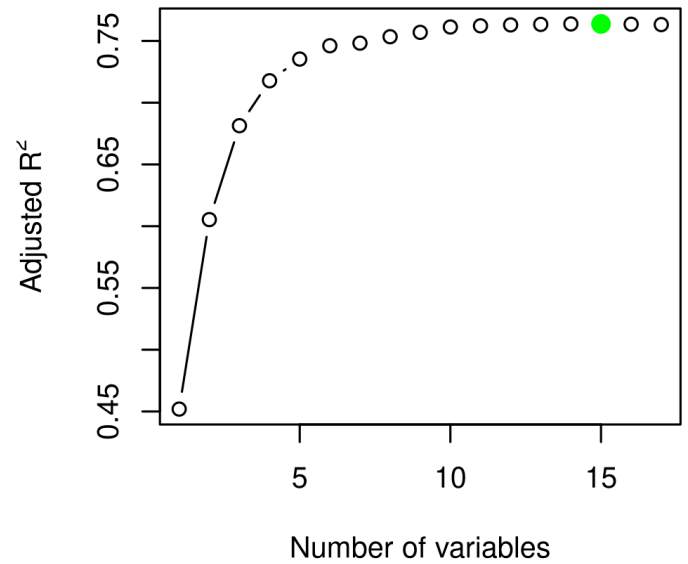
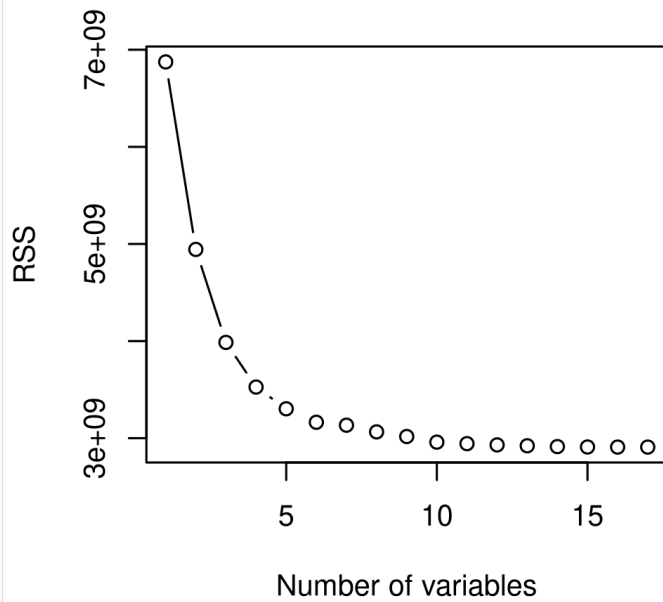
```
## Subset selection object
## Call: regsubsets.formula(Outstate ~ ., data = colDat, nvmax = 20)
## 17 Variables (and intercept)
##              Forced in Forced out
## PrivateYes      FALSE      FALSE
## Apps            FALSE      FALSE
## Accept          FALSE      FALSE
## Enroll          FALSE      FALSE
## Top10perc       FALSE      FALSE
## Top25perc       FALSE      FALSE
## F.Undergrad     FALSE      FALSE
## P.Undergrad     FALSE      FALSE
## Room.Board      FALSE      FALSE
## Books           FALSE      FALSE
## Personal        FALSE      FALSE
## PhD             FALSE      FALSE
## Terminal        FALSE      FALSE
## S.F.Ratio       FALSE      FALSE
## perc.alumni     FALSE      FALSE
## Expend          FALSE      FALSE
## Grad.Rate       FALSE      FALSE
## 1 subsets of each size up to 17
## Selection Algorithm: exhaustive
##      PrivateYes Apps Accept Enroll Top10perc Top25perc F.Undergrad
## 1  ( 1 )      ""      ""      ""      ""      ""      ""      ""
## 2  ( 1 )      ""      ""      ""      ""      ""      ""      ""
## 3  ( 1 )      ""      ""      ""      ""      ""      ""      ""
## 4  ( 1 )      ""      ""      ""      ""      ""      ""      ""
## 5  ( 1 )      ""      ""      ""      ""      ""      ""      ""
## 6  ( 1 )      ""      ""      ""      ""      ""      ""      ""
## 7  ( 1 )      ""      ""      ""      ""      ""      ""      ""
## 8  ( 1 )      ""      ""      *      ""      ""      ""      ""
## 9  ( 1 )      ""      *      *      ""      ""      ""      ""
## 10 ( 1 )      ""      *      *      ""      ""      *      ""
## 11 ( 1 )      ""      *      *      ""      *      ""      ""
## 12 ( 1 )      ""      *      *      ""      *      ""      ""
## 13 ( 1 )      ""      *      *      *      *      ""      ""
## 14 ( 1 )      ""      *      *      *      *      ""      ""
## 15 ( 1 )      ""      *      *      *      *      ""      ""
## 16 ( 1 )      ""      *      *      *      *      *      ""
## 17 ( 1 )      ""      *      *      *      *      *      ""
##      P.Undergrad Room.Board Books Personal PhD Terminal S.F.Ratio
## 1  ( 1 )      ""      ""      ""      ""      ""      ""      ""
## 2  ( 1 )      ""      ""      ""      ""      ""      ""      ""
## 3  ( 1 )      ""      *      ""      ""      ""      ""      ""
## 4  ( 1 )      ""      *      ""      ""      ""      ""      ""
## 5  ( 1 )      ""      *      ""      ""      *      ""      ""
## 6  ( 1 )      ""      *      ""      ""      *      *      ""
## 7  ( 1 )      ""      *      ""      *      *      *      ""
## 8  ( 1 )      ""      *      ""      ""      *      *      ""
## 9  ( 1 )      ""      *      ""      ""      *      *      ""
## 10 ( 1 )      ""      *      ""      ""      *      *      ""
## 11 ( 1 )      ""      *      ""      *      *      *      ""
## 12 ( 1 )      ""      *      ""      *      *      *      ""
## 13 ( 1 )      ""      *      ""      *      *      *      *
## 14 ( 1 )      ""      *      ""      *      *      *      *
## 15 ( 1 )      ""      *      *      *      *      *      *
## 16 ( 1 )      ""      *      *      *      *      *      *
## 17 ( 1 )      ""      *      *      *      *      *      *
##      perc.alumni Expend Grad.Rate
## 1  ( 1 )      ""      *      ""
## 2  ( 1 )      ""      *      ""
## 3  ( 1 )      ""      *      ""
## 4  ( 1 )      *      *      ""
## 5  ( 1 )      *      *      ""
## 6  ( 1 )      *      *      *
## 7  ( 1 )      *      *      *
## 8  ( 1 )      *      *      *
## 9  ( 1 )      *      *      *
## 10 ( 1 )      *      *      *
## 11 ( 1 )      *      *      *
## 12 ( 1 )      *      *      *
## 13 ( 1 )      *      *      *
## 14 ( 1 )      *      *      *
## 15 ( 1 )      *      *      *
## 16 ( 1 )      *      *      *
## 17 ( 1 )      *      *      *
```

We might want to take a look at these.

```
par(mfrow=c(2,2))
plot(modelFit)
plot(modelFit,scale='Cp')
plot(modelFit,scale='r2')
plot(modelFit,scale='adjr2')
```



plotLEAP(modelFit %>% summary)



- So we like 14 variables, namely

```
coefficients(modelFit,id=14) %>% print
```

```
## (Intercept) PrivateYes Apps Accept Enroll
## -1.817040e+03 2.256946e+03 -2.999022e-01 8.023519e-01 -5.372545e-01
## Top10perc F.Undergrad Room.Board Personal PhD
## 2.365529e+01 -9.569936e-02 8.741819e-01 -2.478418e-01 1.269506e+01
## Terminal S.F.Ratio perc.alumni Expend Grad.Rate
## 2.297296e+01 -4.700560e+01 4.195006e+01 2.003912e-01 2.383197e+01
```

- But five seems like a better bet.

```
coefficients(modelFit,id=5)
```

```
## (Intercept) PrivateYes Room.Board PhD perc.alumni
## -2864.6325619 2936.7416766 1.0677573 40.5334088 61.3147684
## Expend
## 0.2253945
```

b) GAM

```
library(gam)
```

```
fit = gam(Outstate ~ Private+s(Apps,3)+Accept+Enroll+
  Top10perc+F.Undergrad+Room.Board+
  Personal+PhD+Terminal+S.F.Ratio+
  perc.alumni+Expend+Grad.Rate
  , data = colDat)
```

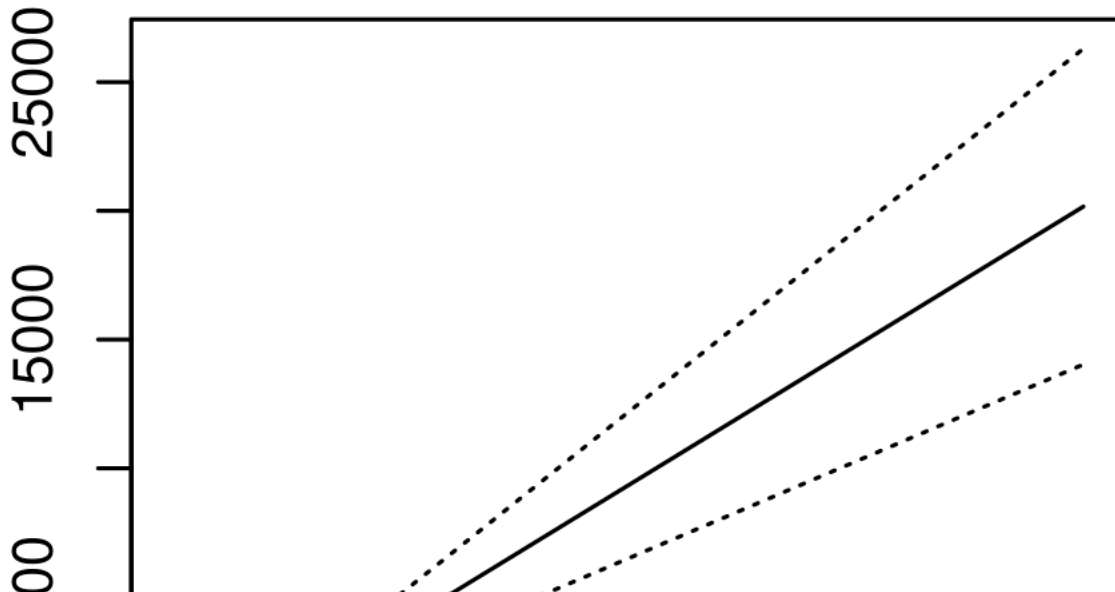
```
## Warning in model.matrix.default(mt, mf, contrasts): non-list contrasts argument
## ignored
```

```
fit2 = gam(Outstate ~ Private+s(Room.Board,2)+s(PhD,3)+s(perc.alumni)+Expend
  , data = colDat)
```

```
## Warning in model.matrix.default(mt, mf, contrasts): non-list contrasts argument
## ignored
```

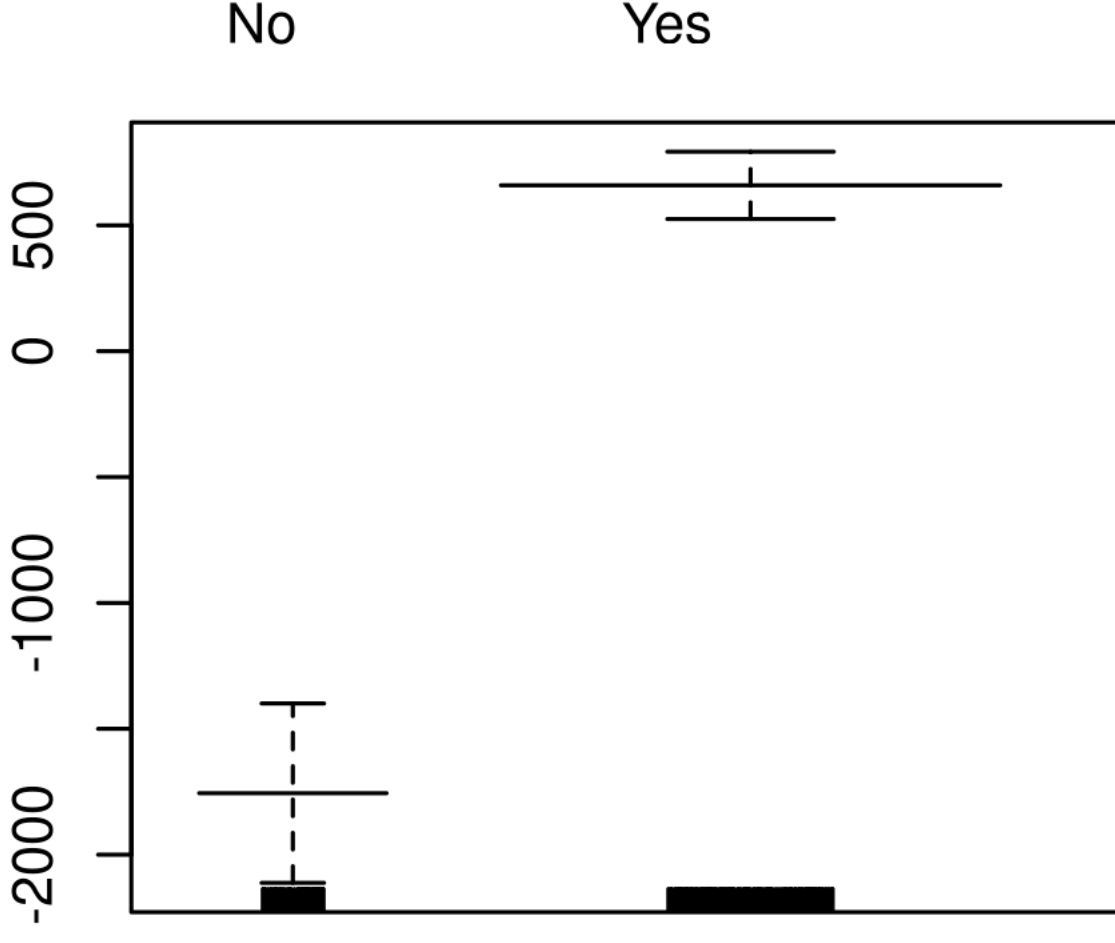
```
par(mfrow=c(2,2))
plot(fit,se=TRUE)
```

partial for Accept

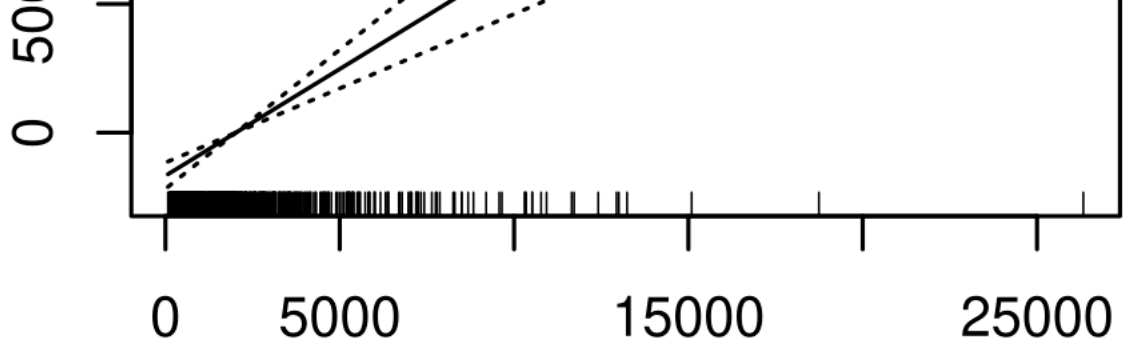


partial for Enroll

partial for Private

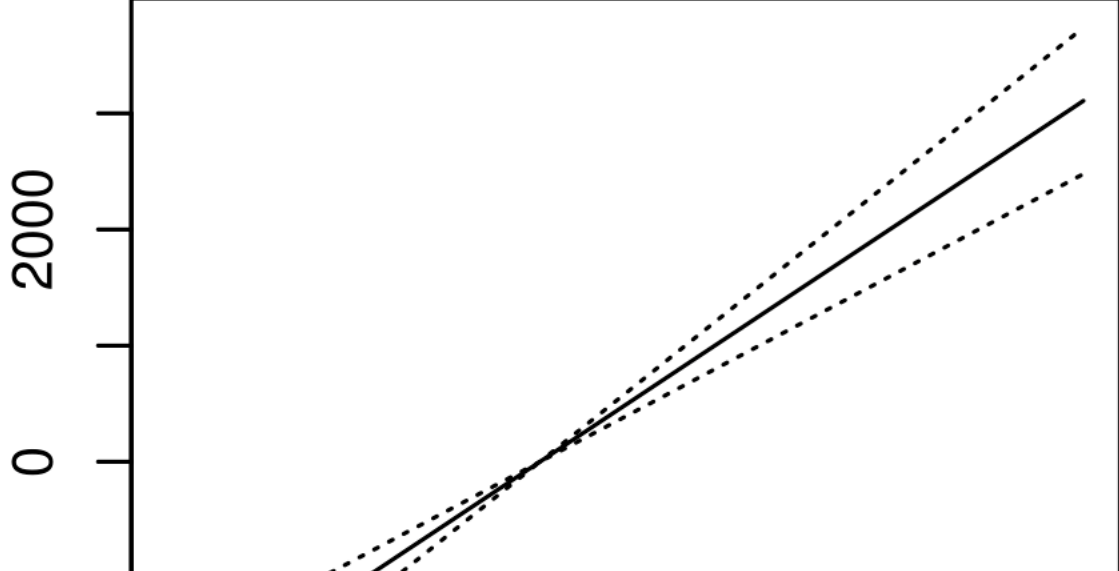


s(Apps, 3)



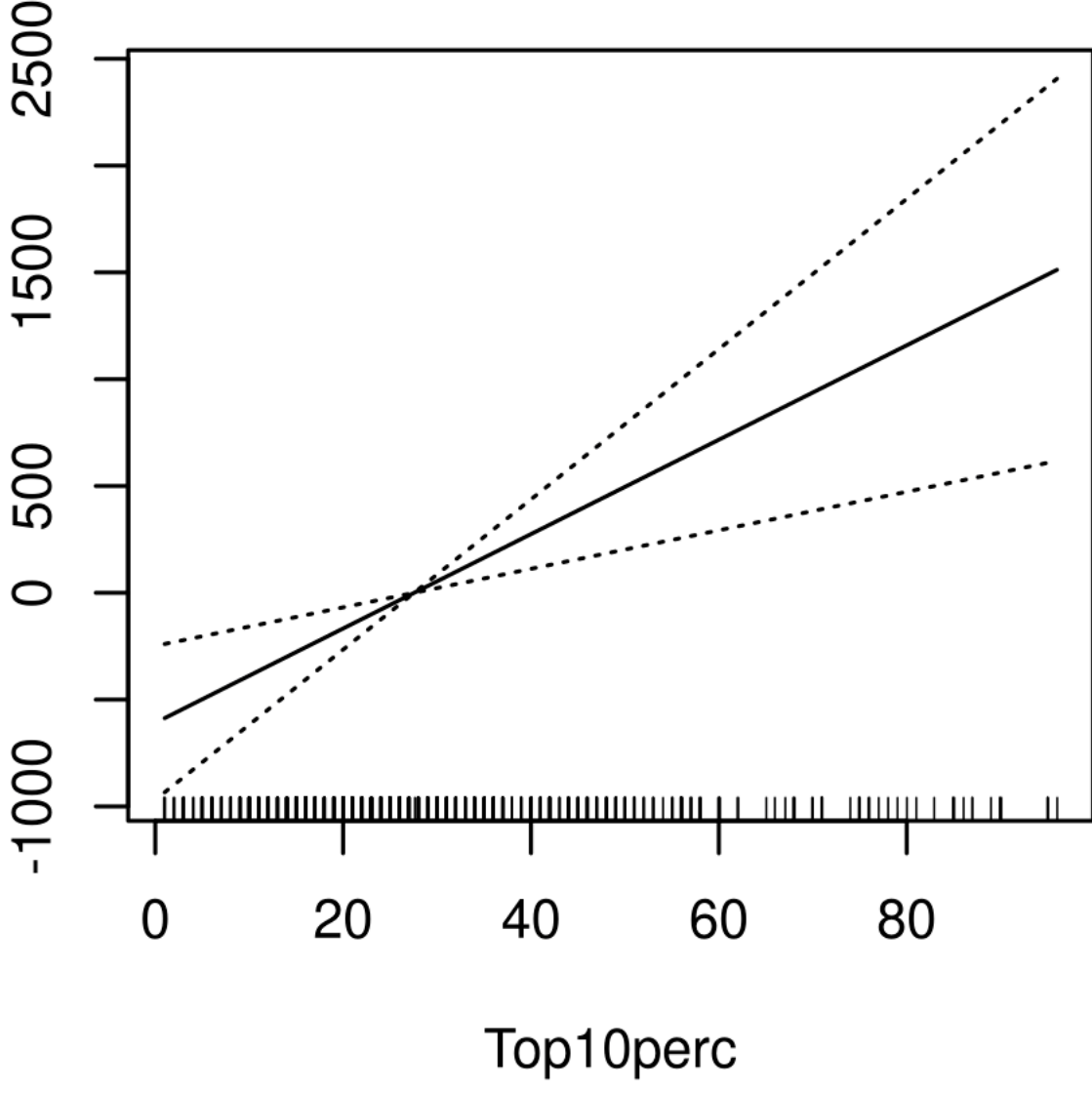
Accept

partial for Room.Board

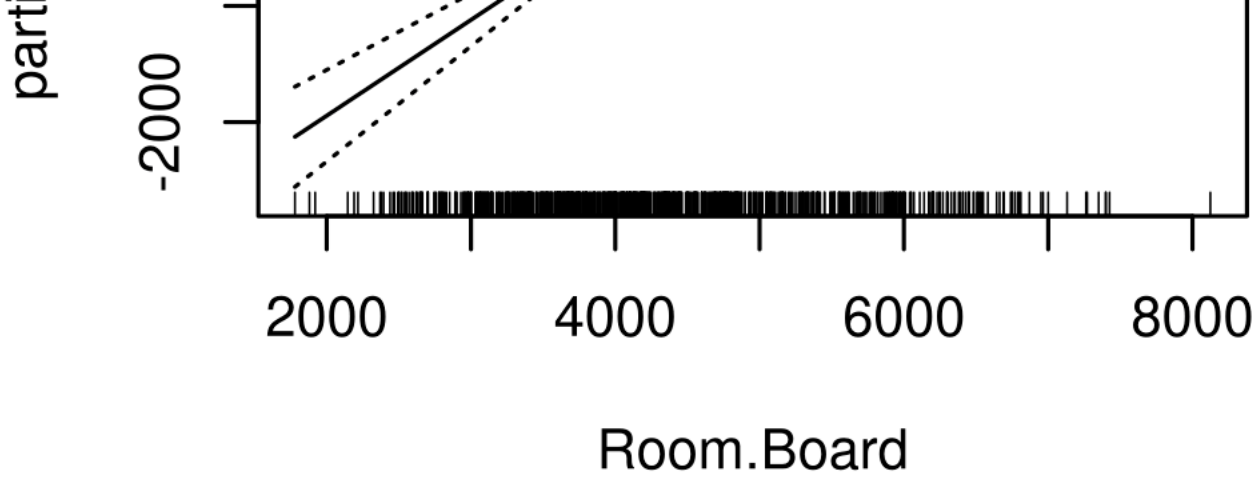


partial for Personal

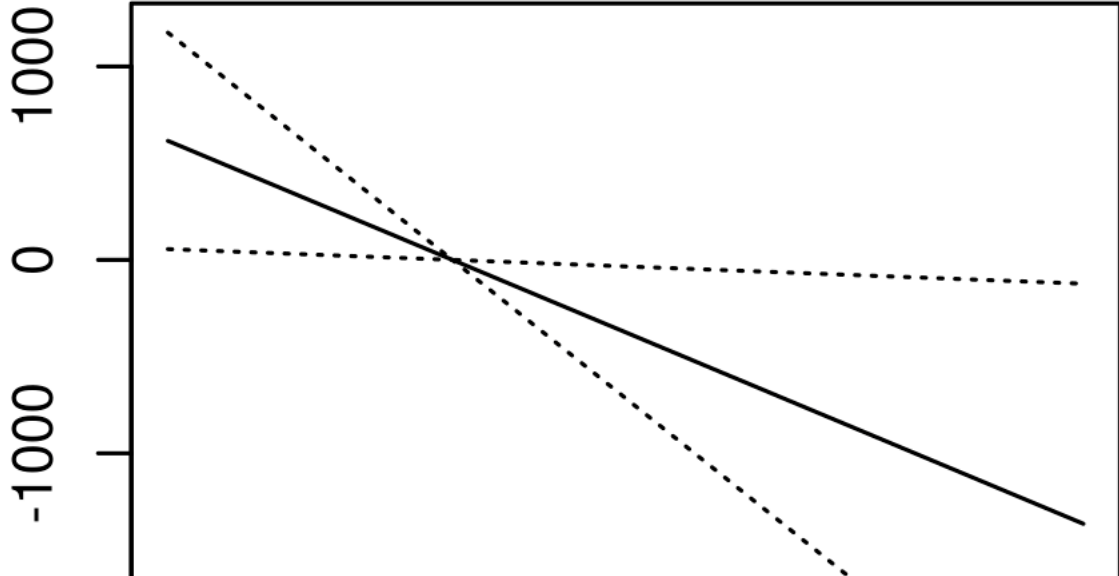
partial for Top10perc



partial for F.Undergrad

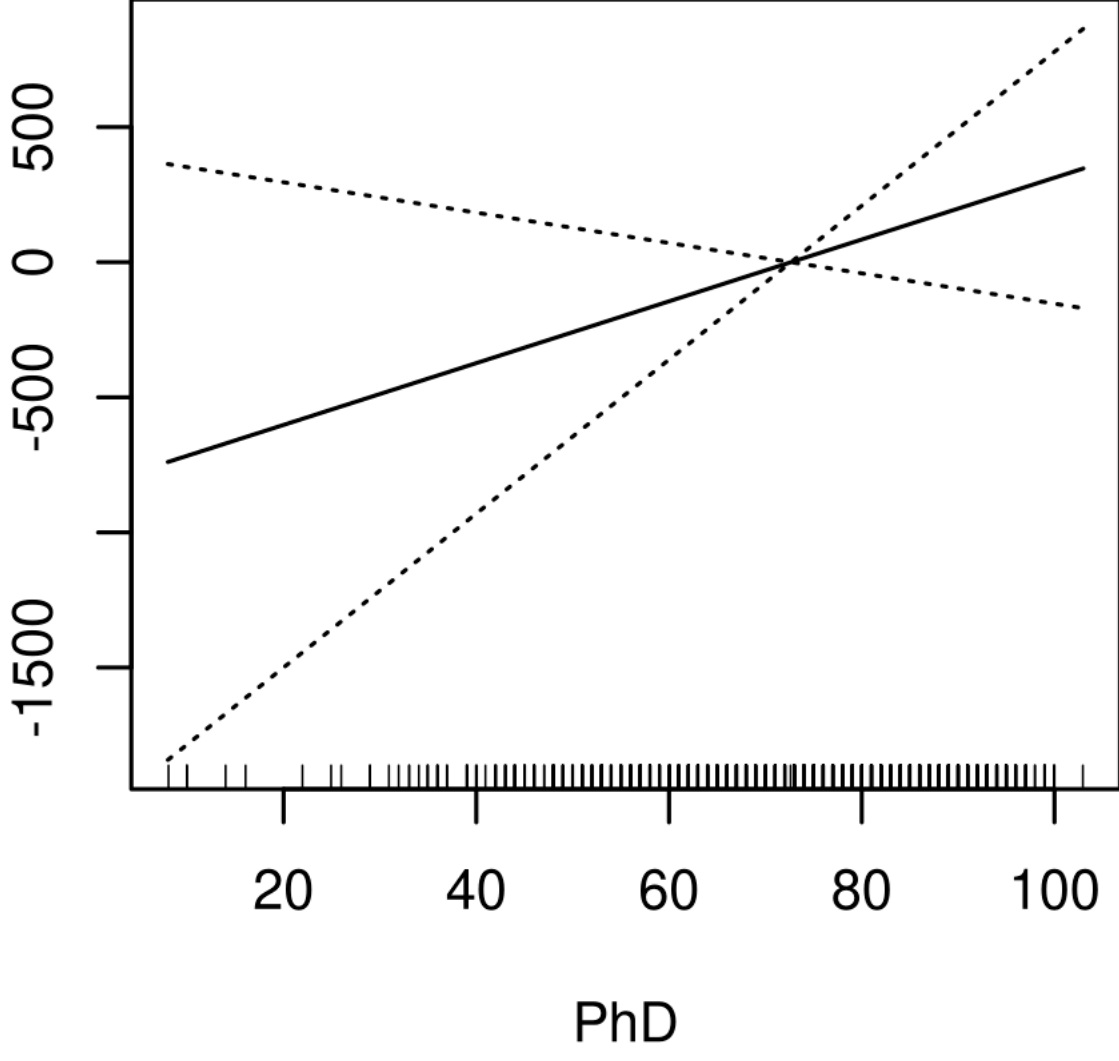


partial for S.F.Ratio



partial for perc.alumni

partial for PhD



partial for Terminal

pa

-2000

10

20

30

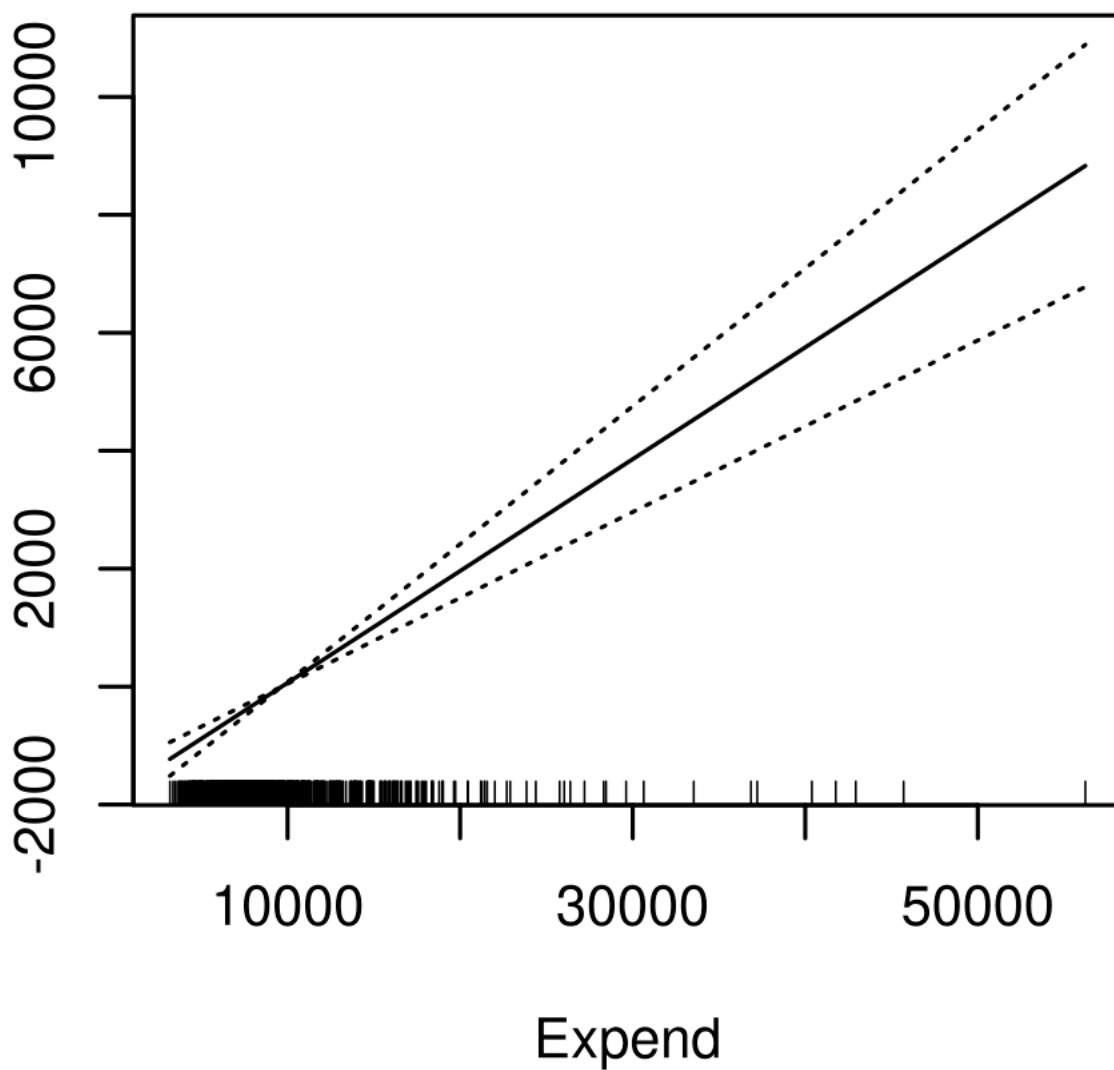
40

S.F.Ratio

par

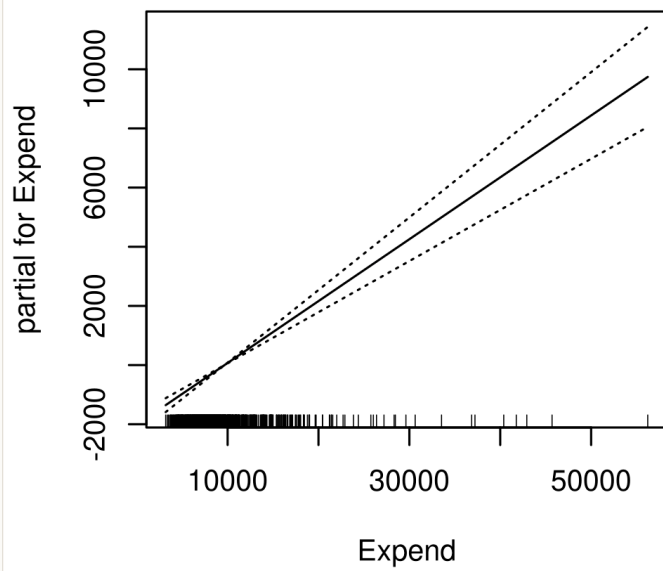


partial for Expend



partial for Grad.Rate

```
par(mfrow=c(2,2))  
plot(fit2,se=TRUE)
```



```
pred <- predict(fit, test_set)
mse.error <- mean((test_set$Outstate - pred)^2)
mse.error %>% print
```

```
## [1] 3691891
```

```
gam.tss = mean((test_set$Outstate - mean(test_set$Outstate))^2)
test.rss = 1 - mse.error/gam.tss
test.rss %>% print
```

```
## [1] 0.7731239
```

```
pred2 <- predict(fit2, test_set)
mse.error2 <- mean((test_set$Outstate - pred2)^2)
mse.error2 %>% print
```

```
## [1] 4121902
```

```
gam.tss2 = mean((test_set$Outstate - mean(test_set$Outstate))^2)
test.rss2 = 1 - mse.error2/gam.tss2
test.rss2 %>% print
```

```
## [1] 0.7466987
```

This is pretty good model, all told.

d) Summary

```
summary(fit) %>% print
```

```
##
## Call: gam(formula = Outstate ~ Private + s(Apps, 3) + Accept + Enroll +
##   Top10perc + F.Undergrad + Room.Board + Personal + PhD + Terminal +
##   S.F.Ratio + perc.alumni + Expend + Grad.Rate, data = colDat)
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -6641.083 -1262.806   -5.698  1270.911  9965.901
##
## (Dispersion Parameter for gaussian family taken to be 3749048)
##
##      Null Deviance: 12559297426 on 776 degrees of freedom
## Residual Deviance: 2849276343 on 760 degrees of freedom
## AIC: 13985.3
##
## Number of Local Scoring Iterations: 2
##
## Anova for Parametric Effects
##      Df      Sum Sq    Mean Sq  F value    Pr(>F)
## Private      1  4034912907 4034912907 1076.250 < 2.2e-16 ***
## s(Apps, 3)    1  1344548030 1344548030  358.637 < 2.2e-16 ***
## Accept       1   90544274   90544274   24.151 1.091e-06 ***
## Enroll       1  144471570   144471570   38.535 8.838e-10 ***
## Top10perc    1 1802244831 1802244831  480.721 < 2.2e-16 ***
## F.Undergrad  1   45230645   45230645   12.065 0.0005430 ***
## Room.Board   1 1110285773 1110285773  296.151 < 2.2e-16 ***
## Personal     1   47886988   47886988   12.773 0.0003738 ***
## PhD          1  220249039  220249039   58.748 5.476e-14 ***
## Terminal     1   66366007   66366007   17.702 2.892e-05 ***
## S.F.Ratio    1  190811028  190811028   50.896 2.274e-12 ***
## perc.alumni  1  225293653  225293653   60.094 2.904e-14 ***
## Expend       1  258162295  258162295   68.861 4.805e-16 ***
## Grad.Rate    1   57947219   57947219   15.457 9.214e-05 ***
## Residuals    760 2849276343    3749048
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Anova for Nonparametric Effects
##      Npar Df Npar F      Pr(F)
## (Intercept)
## Private
## s(Apps, 3)      2  8.571 0.0002085 ***
## Accept
## Enroll
## Top10perc
## F.Undergrad
## Room.Board
## Personal
## PhD
## Terminal
## S.F.Ratio
## perc.alumni
## Expend
## Grad.Rate
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



```
summary(fit2) %>% print
```

```
##
## Call: gam(formula = Outstate ~ Private + s(Room.Board, 2) + s(PhD,
##      3) + s(perc.alumni) + Expend, data = colDat)
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -8676.030 -1345.678   -8.409  1265.524  9590.459
##
## (Dispersion Parameter for gaussian family taken to be 4175193)
##
## Null Deviance: 12559297426 on 776 degrees of freedom
## Residual Deviance: 3194023899 on 765.0002 degrees of freedom
## AIC: 14064.05
##
## Number of Local Scoring Iterations: 2
##
## Anova for Parametric Effects
##              Df      Sum Sq   Mean Sq F value    Pr(>F)
## Private         1 3751107814 3751107814   898.43 < 2.2e-16 ***
## s(Room.Board, 2) 1 2913770756 2913770756   697.88 < 2.2e-16 ***
## s(PhD, 3)        1 1149711330 1149711330   275.37 < 2.2e-16 ***
## s(perc.alumni)   1  556759894  556759894   133.35 < 2.2e-16 ***
## Expend          1  554812125  554812125   132.88 < 2.2e-16 ***
## Residuals       765 3194023899   4175193
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Anova for Nonparametric Effects
##              Npar Df Npar F      Pr(F)
## (Intercept)
## Private
## s(Room.Board, 2)      1 4.9853 0.0258517 *
## s(PhD, 3)            2 9.1614 0.0001171 ***
## s(perc.alumni)       3 0.8726 0.4548496
## Expend
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

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